A quenched invariance principle for reflected walks with almost-everywhere line connectivity

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Abstract

We extend the scale-free cell-environment argument to random walk reflected off of infinity. Bounded Euclidean sets may contain infinitely many cells, while the adjacency graph is locally finite. The geometric assumption is finite-path connectedness along every segment on almost every horizontal and almost every vertical line; the exceptional parameters are line offsets. Spatial rectangle restrictions need not be connected. We instead prove that every graph component meets the prescribed spatial boundary and solve the full finite-energy Dirichlet problem componentwise, retaining free conditions at finite spatial ends. We construct harmonic coordinates with a uniformly sublinear corrector, exclude visits to spatial infinity, construct exponential and exact area clocks, and prove the harmonic-coordinate martingale and its full bracket through reflection times. Temporal mass transport and regeneration yield a quenched Brownian limit with deterministic positive-definite covariance. No joint ergodicity assertion for an auxiliary dyadic grid is used. The proof includes all spatial and process-level arguments, with the external probabilistic theorems identified explicitly.

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1 Environment, boundary conditions, and the main theorem

We identify $\mathbb{C}$ with $\mathbb{R}^2$. All graph energies below sum over *unoriented* edges, once per edge. Write $\mathrm{d}f(H,H') = f(H') - f(H)$ for a gradient and $\operatorname{osc}_V f = \sup_{H,H'\in V} |f(H) - f(H')|$ for oscillation, also for vector-valued functions. Similarities mean translations and positive dilations, without rotations.

Definition 1.1 (Environment and spatial connectivity)**.** A generalized cell environment is a countable family $\mathcal{H}$ of compact connected subsets of $\mathbb{C}$, with a graph adjacency relation and conductances, such that:

(i) every cell has nonempty interior and Lebesgue-null boundary; the interiors of distinct cells are disjoint and $\bigcup_{H\in\mathcal{H}} H = \mathbb{C}$;

(ii) the adjacency graph is connected and locally finite, and $H \sim H'$ implies $H \neq H'$ and $H \cap H' \neq \varnothing$;

(iii) each adjacent pair has a symmetric conductance $c(H,H') \in (0,\infty)$.

For a set $A \subset \mathbb{C}$, $\mathcal{H}(A)$ is the induced subgraph on the cells meeting A. We assume that there exist Lebesgue-measurable null sets $N_h, N_v \subset \mathbb{R}$ such that

$$\begin{aligned} &\mathcal{H}([a,b]\times\{y\}) \text{ is connected for every } a<b \quad \text{whenever } y \notin N_h,\\ &\mathcal{H}(\{x\}\times[a,b]) \text{ is connected for every } a<b \quad \text{whenever } x \notin N_v. \end{aligned} \tag{AE-LC}$$

Connected means connected by finite graph paths, not just connected after adding accumulation points. The sets N_h, N_v may depend on the environment but work simultaneously for all real segment endpoints. They concern line offsets, not line directions. The good-offset sets need not be open or contain any rational points; only their conullness is used. The relevant subgraphs may be infinite.

This weakens the every-line condition of [1, Theorem 1.16]. It is a hypothesis, not a consequence of the moment condition. We assume neither connectivity nor finiteness of a spatial rectangle restriction or its boundary. The exceptional offsets are used only in pathwise line selections; no measurable choice of their null sets as a function of the environment is needed. The environment law is understood to be supported on a measurable probability-one set where (AE-LC) holds. This property is similarity invariant: under $z \mapsto C(z-u)$, the exceptional offsets can be transformed to $C(N_h - \operatorname{Im} u)$ and $C(N_v - \operatorname{Re} u)$.

Write

$$\begin{aligned} a_H &= \operatorname{Area}(H), \qquad d_H = \operatorname{diam}(H), \qquad b_H = \frac{1}{a_H}\int_H z\,dz,\\ \pi(H) &= \sum_{H'\sim H} c(H,H'), \qquad \pi^*(H) = \sum_{H'\sim H} c(H,H')^{-1}. \end{aligned} \tag{1.1}$$

The centroid b_H need not lie in a nonconvex cell, but

$$|b_H - z| \le d_H \quad (z\in H), \qquad |b_H - b_{H'}| \le d_H + d_{H'} \quad (H\sim H'). \tag{1.2}$$

Let H_z denote the unique cell containing z when it is unique. Rooted cell functionals, including the density in the moment condition below, are defined to be zero at boundary roots. The boundary transport below proves that the original root is almost surely not exceptional.

The probability law of $\mathcal{H}$ satisfies *mass transport modulo scaling* if

$$\mathbf{E}\Big[\int_{\mathbb{C}} T(\mathcal{H},0,z)\,dz\Big] = \mathbf{E}\Big[\int_{\mathbb{C}} T(\mathcal{H},z,0)\,dz\Big] \tag{1.3}$$

for each nonnegative measurable T with

$$T(C(\mathcal{H}-u),C(w-u),C(z-u)) = C^{-2}T(\mathcal{H},w,z). \tag{1.4}$$

This is the mass-transport formulation of [1, Definition 1.2]. The moment assumption is

$$M := \mathbf{E}\left[\frac{d_{H_0}^2}{a_{H_0}}\big(\pi(H_0)+\pi^*(H_0)\big)\right] < \infty. \tag{FE}$$

The origin lies off every cell boundary almost surely under (1.3); this is verified below.

For a graph G and real- or $\mathbb{R}^2$-valued functions, set

$$\mathcal{E}_G(f,g) = \sum_{\{H,H'\}\in E(G)} c(H,H')\big(f(H')-f(H)\big)\cdot\big(g(H')-g(H)\big), \qquad \mathcal{E}_G(f) = \mathcal{E}_G(f,f). \tag{1.5}$$

For a bounded closed axis-parallel rectangle Q with nonempty interior put

$$G_Q = \mathcal{H}(Q), \quad A_Q = \mathcal{H}(\partial Q), \quad \mathcal{V}_Q = \{u : V(G_Q)\to\mathbb{R}^2 : u|_{A_Q} = 0,\ \mathcal{E}_{G_Q}(u) < \infty\}. \tag{1.6}$$

In particular, elements of $\mathcal{V}_Q$ are not required to vanish near accumulation points or to have finite vertex support.

Theorem 1.2 (Harmonic coordinates)**.** *Assume Definition 1.1,* (AE-LC)*,* (1.3)*, and* (FE)*. There is a measurable function $\Phi = \Phi_{\mathcal{H}} : \mathcal{H}\to\mathbb{R}^2$, normalized by $\Phi(H_0) = 0$, with the following properties almost surely.*

(a) For every bounded closed axis-parallel rectangle Q, $\mathcal{E}_{G_Q}(\Phi) < \infty$ and

$$\mathcal{E}_{G_Q}(\Phi,u) = 0 \qquad \text{for every } u\in\mathcal{V}_Q. \tag{1.7}$$

Equivalently, $\Phi|_{G_Q}$ uniquely minimizes the full graph energy among functions with boundary trace $\Phi|_{A_Q}$. In particular, Φ is discrete harmonic at every vertex.

(b) The gradient has finite specific energy:

$$\mathbf{E}\left[\frac{1}{2a_{H_0}}\sum_{H\sim H_0} c(H_0,H)|\Phi(H)-\Phi(H_0)|^2\right] < \infty. \tag{1.8}$$

Its gradient construction commutes with translations and positive dilations. Where H_u is uniquely defined, the normalized covariance rule is

$$\Phi_{C(\mathcal{H}-u)}(C(H-u)) = C\big(\Phi_{\mathcal{H}}(H)-\Phi_{\mathcal{H}}(H_u)\big). \tag{1.9}$$

(c) The corrector is uniformly sublinear:

$$\lim_{R\to\infty}\frac{1}{R}\sup_{H\in\mathcal{H}(\overline{B}_R(0))}|\Phi(H)-b_H| = 0. \tag{1.10}$$

The same statement holds with b_H replaced by any choice of a point of H.

(d) If ϕ_m denotes the blockwise energy-minimizing interpolation constructed in Section 4, then

$$\phi_m - \phi_m(H_0) \longrightarrow \Phi \tag{1.11}$$

in probability, uniformly on each fixed finite graph neighborhood of H_0. The gradients converge in specific energy. A deterministic subsequence converges almost surely in energy on every bounded spatial patch and pointwise at every vertex.

The final Φ does not depend on the auxiliary dyadic system used to define the ϕ_m.

Ergodicity modulo scaling is not needed for this harmonic-coordinate theorem. It makes the eventual spatial energy bound deterministic. The stronger conclusion needed here is that the approximation errors have uniformly small spatial averages; the maximal inequality below gives this directly.

For the process-level conclusions we also assume that every measurable event invariant under translations and positive dilations of the environment has probability zero or one. Fix a measurable similarity-covariant representative $z_H \in H$; for instance choose a point nearest to b_H, resolving ties lexicographically. The probabilistic root remains H_0, not this representative.

Version of the reflected process. We use the path-regular, constructed reflected walk of Gwynne–Sung, with an admissible fast clock. In particular we work with its regular realization of the full reflected form when using the Dirichlet-form theorems. At a nonvertex time the collapsed path is right-continuous in the one-point compactification of the discrete vertex set: each finite set of vertices is avoided on a sufficiently short interval to the right. The theorem below concerns this canonical version and its homeomorphic time changes, not arbitrary modifications at exceptional random times which preserve all fixed-time distributions. This distinction is important for attained return times and simultaneous assertions about ends. In particular we do not infer regularity of arbitrary path versions merely from equality of fixed-time laws, and we do not use a uniqueness claim over a larger unregularized class to obtain these path properties.

Theorem 1.3 (Reflected invariance principle under almost-everywhere line connectivity)**.** *Under Definition 1.1,* (AE-LC)*,* (1.3)*,* (FE)*, and the environment ergodicity just stated, the following hold conditionally on almost every environment.*

(a) The reflected walk admits an area clock X, with transition rates $c(H, H')/a_H$ while at vertices and mean holding time $a_H/\pi(H)$. It is defined for all positive times, returns to every vertex, and is a homeomorphic time change of an admissible fast-clock reflected walk.

(b) The vertex-valued spatial path z_{X_t} has a unique extension to a càdlàg path Z which is continuous at every end-valued time. Its only jumps are ordinary graph-edge jumps. It never hits a spatial-infinity end.

(c) The harmonic-coordinate path $M_t = \Phi(X_t)$ has the analogous extension and is a local martingale, including through reflection times. Its predictable bracket is

$$\langle M^i, M^j\rangle_t = \int_0^t \Gamma_{ij}(\mathcal{H}, X_s)\, ds, \quad \Gamma(\mathcal{H}, H) = \frac{1}{a_H} \sum_{H' \sim H} c(H, H')\Delta\Phi\, \Delta\Phi^{\mathsf{T}}, \tag{1.12}$$

where $\Delta\Phi = \Phi(H') - \Phi(H)$ and the integrand is set to zero at end-valued times.

(d) With $\Sigma = \mathbf{E}[\Gamma(\mathcal{H}, H_0)]$, Σ *is finite and positive definite. If* $\widetilde{Z}$ *linearly interpolates ordinary jumps over their preceding holding intervals and uses the limiting spatial values at reflection times, then*

$$\mathcal{L}\big((\varepsilon \widetilde{Z}_{t/\varepsilon^2})_{t\geq 0} \mid \mathcal{H}\big) \implies \mathcal{L}\big((B_t)_{t\geq 0}\big) \quad \text{in } C([0,\infty), \mathbb{R}^2) \tag{1.13}$$

with the topology of uniform convergence on compact time intervals. Here B *starts at zero and* $\mathbf{E}[B_t B_t^{\mathsf{T}}] = t\Sigma$.

(e) The same Brownian limit holds for the same ordered reflected path with exact *holding times* $a_H/\pi(H)$*, and for any other measurable representatives in the cells.*

The convergence holds from each fixed starting vertex. Uniformity over starting vertices depending on ε *is not asserted.*

An end is at spatial infinity when its nested components C_n outside a finite graph exhaustion satisfy $\inf_{H\in C_n} \operatorname{dist}(0, H) \to \infty$. Existence or uniqueness of such an end is not assumed. The theorem does not assert that every graph end admits a single deterministic spatial image. The pathwise extension at an end-valued time is all that is needed for (1.13).

1.1 External inputs and proof order

All nonstandard spatial estimates and the corrector are proved below. From [2] we use the constructed reflected process, its finite-target energy-minimizing hitting laws, vertex-valued strong Markov property, canonical fixed-vertex path regularity, end labeling, and identification with the full reflected form (Theorem 1.6, Lemmas 3.10–3.13, Proposition 3.15, and Proposition 5.3). We apply these to the regular version specified above.

The analytic external inputs are the regular symmetric Dirichlet form–Hunt-process correspondence, Beurling–Deny and Fukushima decompositions, and the full-quasi-support time-change theorem [3, 4]. The bounded global form-domain case of [5, Theorem 2.7 and Remark 2.8(i)] identifies weak harmonicity with the stopped martingale property on relatively compact open subsets. Section 12 constructs the regular space and checks the boundary energy and reference measure; Section 13 checks the harmonicity tests. For the final stochastic limit we use [7, Theorem 2.1(ii)] after bounded-jump localization. These external results are not assumed to have Lean proofs.

The order is spatial estimates, full-energy minimization on boundary-anchored components, corrector sublinearity, avoidance of spatial infinity, finite-cycle occupation and clocks, reflected martingales, temporal averaging, and the quenched limit. Ordinary random-walk recurrence is never assumed. The almost-everywhere line hypothesis is consumed only in the good-line oscillation argument; it is not needed for the recurrence estimate or for anchoring components at spatial boundaries.

2 Spatial tools without a joint ergodicity assertion

The distinction between conditional block averaging and joint grid ergodicity is explained in Appendix A. We use no assertion that $\mathcal{H}$ together with an independent uniform dyadic system is ergodic.

2.1 Null boundaries, large cells, and dyadic blocks

Mass transport implies that the origin is not on a cell boundary. Indeed, apply it to

$$T(\mathcal{H}, w, z) = \sum_{H \in \mathcal{H}} \frac{\mathbf{1}_{\{w \in \partial H\}} \mathbf{1}_{\{z \in H\}}}{a_H}.$$

The incoming integral is zero and the outgoing integral counts the cell boundaries containing the origin. Thus that count vanishes almost surely. Rooted cell functionals below are set to zero on the exceptional set of boundary roots; this gives an exactly covariant convention.

Lemma 2.1 (Spatial finiteness above a scale)**.** *Almost surely, for every $K > 0$,*

$$\#\{H : \operatorname{dist}(0, H) \le K d_H\} < \infty. \tag{2.1}$$

Consequently, for every finite R and every $\delta > 0$, only finitely many cells meeting $\overline{B}_R(0)$ have diameter at least δ, and

$$D_R := \sup_{H \in \mathcal{H}(\overline{B}_R(0))} d_H < \infty, \qquad D_R/R \longrightarrow 0. \tag{2.2}$$

Proof. Every vertex has a neighbor, so $\pi(H) + \pi^*(H) \ge 2$ and $\mathbf{E}[d_{H_0}^2/a_{H_0}] < \infty$. Write $H^{[K]} = \{z : \operatorname{dist}(z, H) \le K d_H\}$. The transport

$$T_K(\mathcal{H}, w, z) = a_{H_w}^{-1} \mathbf{1}_{\{z \in H_w^{[K]}\}}$$

has expected outgoing mass at most $\pi(K+1)^2 \mathbf{E}[d_{H_0}^2/a_{H_0}]$ and incoming mass equal to the count in (2.1). Here and below transports are zero at an ambiguous source cell; ambiguous target roots do not affect the asserted almost-sure identities. Intersect the resulting events for positive integer K.

If H meets $\overline{B}_R(0)$ and $d_H \ge \delta$, it is counted with $K \ge R/\delta$. The asserted finiteness follows; taking $\delta = 1$ gives $D_R < \infty$. If $d_H \ge \varepsilon R$ and H meets $\overline{B}_R(0)$, it is counted with $K \ge 1/\varepsilon$. This is one fixed finite collection as R varies. Its diameters are fixed and finite, so for all sufficiently large R none exceeds εR. Intersect over rational $\varepsilon > 0$. □

We use the uniform dyadic system of [1, Section 2.1 and Lemma 2.1]. Its side lengths are 2^{s+k}, $k \in \mathbb{Z}$, with s uniform on $[0, 1)$; at a fixed level the origin position is uniform in its square, and successive parents are chosen uniformly from the four possibilities. Its law is invariant under every deterministic translation and positive dilation. Almost surely any two squares have a common ancestor, and the origin-containing ancestor chain exhausts $\mathbb{C}$.

Here is the block index used below. For $S \in \mathcal{D}$, let $S^{(j)}$ be its jth ancestor and set

$$D(S) = \sup_{H \cap S \neq \varnothing} d_H, \quad a(S) = \sup_{j \ge 0} \frac{D(S^{(j)})}{\ell(S^{(j)})}, \quad b(S) = a(S)^{-1}, \quad \kappa(S) = \sum_{j \ge 0} 4^{-j} b(S^{(j)}). \tag{2.3}$$

The symbol $b(S)$ in this display is an index; b_H always denotes a cell centroid. By (2.2), the quantities in (2.3) are finite and positive. If $P = S^{(1)}$, then

$$b(S) \le b(P) \le 2b(S), \quad \tfrac{4}{3} b(S) < \kappa(S) \le 2b(S), \quad \kappa(P) = 4\big(\kappa(S) - b(S)\big) > \kappa(S). \tag{2.4}$$

Indeed $D(S) \le D(P)$ and $\ell(P) = 2\ell(S)$ give the first inequalities; the geometric series gives the second, strictly because $b(S^{(j)}) \to \infty$. The displayed identity gives the last inequality. Along

ancestors $\kappa \to \infty$. Along a decreasing chain containing a point $z \in H$ one has $D(S) \geq d_H$ and hence $\kappa(S) \leq 2\ell(S)/d_H \to 0$.

For $m > 0$ let

$$\mathcal{S}_m = \{S \in \mathcal{D} : \kappa(S) \leq m < \kappa(S^{(1)})\}. \tag{2.5}$$

These squares have disjoint interiors and their closures cover the plane. For the latter assertion, choose any dyadic chain through a given point, including a consistent choice at a dyadic boundary, and use the two limits of κ. A point in the interior of a selected square belongs to no other selected square. Let $S_m(z)$ be this square off the selected boundaries. The partitions are nested in m, commute with similarities, and the blocks through a fixed generic point exhaust the plane. Every origin-containing dyadic square is $S_m(0)$ for all m in a nonempty interval. No number of cells is used in this construction.

2.2 Re-rooting and a maximal inequality

We will sometimes attach one or more independent uniform dyadic systems to $\mathcal{H}$. Mass transport remains valid for a transport which depends covariantly on these marks: average the transport over the marks before applying (1.3). The invariance of their laws gives (1.4) for the averaged transport.

Lemma 2.2 (Conditional block averaging)**.** *Let ω consist of $\mathcal{H}$ and any such independent marks, and let $\mathcal{D}'$ be a further independent uniform dyadic system. Construct $S_m(z)$ using $\mathcal{H}, \mathcal{D}'$. If $F(\omega) \geq 0$ is integrable and scale invariant, then*

$$A_m = \frac{1}{\ell(S_m(0))^2} \int_{S_m(0)} F(\omega - z)\, dz = \mathbf{E}[F \mid \mathcal{G}_m], \tag{2.6}$$

Here $\mathcal{G}_m$ is the completed sigma-field of measurable events of $(\omega, \mathcal{D}')$ whose indicators are unchanged by common scaling and by translating a point of $S_m(0)^\circ$ to the origin. This is interpreted on the set where the selected origin block is unique; selected-block boundary roots have probability zero. Invariance is required wherever the two rooted configurations have unique blocks. The sigma-fields $\mathcal{G}_m$ decrease with m.

Proof. For an arbitrary nonnegative scale-invariant $U(\omega, \mathcal{D}')$, apply marked mass transport to

$$T(\omega, \mathcal{D}', w, z) = \frac{\mathbf{1}_{\{z \in S_m(w)\}}}{\ell(S_m(w))^2} U(\omega - z, \mathcal{D}' - z).$$

The incoming integral is $U(\omega, \mathcal{D}')$, since $z \in S_m(w)$ is equivalent, almost everywhere, to w, z being in the same selected block. Thus the block average of U has the same expectation as U. If $z \in S_m(0)^\circ$, its selected block is the same square before re-centering. A change of variables, and scale invariance of F, therefore leave A_m unchanged by every allowed re-rooting and scaling. Thus A_m is $\mathcal{G}_m$-measurable. Apply the preceding expectation identity to $U = \mathbf{1}_B F$ for $B \in \mathcal{G}_m$. Inside the origin block, $\mathbf{1}_B$ is unchanged, so

$$\mathbf{E}[\mathbf{1}_B A_m] = \mathbf{E}[\mathbf{1}_B F].$$

These are the measurability and testing identities for conditional expectation. Finally an invariant event for the larger block is invariant for its smaller subblock; this gives the decreasing sigma-fields. Completing them under the common law does not change these identities. □

Proposition 2.3 (Spatial maximal inequality)**.** *For every nonnegative integrable scale-invariant $F(\omega)$, the density $\rho_\omega(z) = F(\omega - z)$ is locally integrable almost surely, and*

$$\mathbf{P}\left[\mathcal{M}(\rho) > \lambda\right] \le \frac{512\,\mathbf{E}[F]}{\lambda}, \qquad \mathcal{M}(\rho) := \sup_{R>0} \frac{1}{R^2} \int_{\overline{B}_R(0)} \rho(z)\,dz, \quad \lambda > 0. \tag{2.7}$$

In particular, $\mathcal{M}(\rho) < \infty$ almost surely. The claim is valid with the independent dyadic marks present and does not require a trivial invariant sigma-field.

Proof. For any finite increasing list of positive rational m's, the conditional expectations in (2.6), read in reverse order, are a nonnegative martingale. Doob's weak-L^1 maximal inequality, followed by increasing finite collections exhausting the positive rational parameters, gives

$$\mathbf{P}\left[\sup_{m\in\mathbb{Q}_{>0}} A_m > t\right] \le \mathbf{E}[F]/t. \tag{2.8}$$

This argument involves no continuous-parameter martingale convergence theorem. Every origin dyadic square occurs for an interval of parameters containing a rational. Thus (2.8) bounds the maximal average over the complete origin dyadic chain. Each such square has a finite integral almost surely, and these squares exhaust the plane. This proves local integrability. This conclusion concerns ω alone, so the auxiliary $\mathcal{D}'$ can be integrated out.

On $\{\mathcal{M}(\rho) > \lambda\}$ choose a positive rational R, measurably from ω and independently of $\mathcal{D}'$, for which $R^{-2}\int_{\overline{B}_R(0)} \rho > \lambda$. Such a choice is possible because the ball integrals of a locally integrable density are continuous in $R > 0$. Only after this choice of R do we use the further independent grid $\mathcal{D}'$. Conditional on its phase, choose its level of side length $\ell \in [8R, 16R)$. With conditional probability at least $(3/4)^2 > 1/2$, its origin square contains $\overline{B}_R(0)$. On that event the square average exceeds $\lambda/256$. Consequently

$$\tfrac{1}{2}\mathbf{P}[\mathcal{M}(\rho) > \lambda] \le \mathbf{P}[\text{some origin dyadic average exceeds } \lambda/256] \le 256\mathbf{E}[F]/\lambda.$$

This is (2.7). □

Corollary 2.4 (Local and large-scale energy bounds)**.** *Almost surely,*

$$W(R) := \sum_{H\in\mathcal{H}(\overline{B}_R(0))} d_H^2(\pi(H) + \pi^*(H)) < \infty \quad (R < \infty), \qquad \limsup_{R\to\infty} \frac{W(R)}{R^2} < \infty. \tag{2.9}$$

If the environment law is ergodic modulo translations and positive dilations, the last limsup is deterministic.

Proof. Apply Proposition 2.3 to

$$F_q(\mathcal{H}) = \frac{d_{H_0}^2(\pi(H_0) + \pi^*(H_0))}{a_{H_0}}, \qquad \rho_q(z) = F_q(\mathcal{H} - z).$$

Every cell meeting $\overline{B}_R(0)$ lies in $\overline{B}_{R+D_R}(0)$. Its entire weight is counted by integration of ρ_q there. Hence

$$W(R) \le \int_{\overline{B}_{R+D_R}(0)} \rho_q \le (R + D_R)^2 \mathcal{M}(\rho_q).$$

Use (2.2). Finally the finite limsup in (2.9) is unchanged by a translation of the origin (squeeze between radii $R \pm |u|$) or a common dilation (the weights scale quadratically). Ergodicity makes it deterministic. □

This proves the required spatial energy bounds without deterministic averaging of arbitrary dyadic-marked functionals.

3 Full finite-energy minimization on boundary-anchored components

Definition 3.1 (Boundary anchoring)**.** For a countable graph G and $A \subset V(G)$, the pair (G, A) is boundary anchored if every graph component of G contains a vertex of A. Components and all paths here are graph-theoretic. The set A, the graph, and the number of components may be infinite.

Lemma 3.2 (Full-energy Dirichlet problem)**.** *Let G be a countable locally finite conductance graph, not necessarily connected, and let (G, A) be boundary anchored. If $g : V(G) \to \mathbb{R}^2$ has finite total energy, there is a unique minimizer h of $\mathcal{E}_G(f)$ over all functions with $f|_A = g|_A$. It is characterized by*

$$\mathcal{E}_G(h, u) = 0 \qquad (u|_A = 0,\ \mathcal{E}_G(u) < \infty). \tag{3.1}$$

On each component, every coordinate bound on its boundary trace also bounds that coordinate of h. In particular a common bound on all of A bounds h on all vertices.

Proof. Orient the edges arbitrarily and give edge fields the Hilbert norm $\sum_e c(e)|\theta(e)|^2$. For each component C, select $a_C \in A \cap C$. The gradients of finite-energy functions vanishing on A form a closed linear subspace. Indeed, if $\mathrm{d}u_n$ is Cauchy in that norm, then for each $x \in C$ and a fixed finite path P_x from a_C to x,

$$|u_n(x) - u_k(x)|^2 \le \left(\sum_{e \in P_x} c(e)^{-1} \right) \mathcal{E}_G(u_n - u_k). \tag{3.2}$$

Thus the potentials converge at every vertex to a function zero on A; edgewise convergence identifies its gradient with the Hilbert limit. Project $\mathrm{d}g$ onto this closed subspace and subtract its anchored potential. The resulting h has the required trace and satisfies (3.1). Conversely orthogonality gives

$$\mathcal{E}_G(h + u) = \mathcal{E}_G(h) + \mathcal{E}_G(u).$$

This proves minimality. If equality holds, u is constant on each component, and its value at that component's anchor is zero. Thus the minimizer is unique. Clipping one coordinate, either on one component or on the whole graph, preserves the prescribed trace when the asserted bound holds, and cannot increase energy. Uniqueness proves the maximum principle. The construction uses the one global edge Hilbert space; it does not assume that infinitely many independently chosen minimizers can be assembled with finite energy without a bound. □

Lemma 3.3 (Anchored finite approximation and measurability)**.** *In Lemma 3.2, choose increasing finite sets $V_n \uparrow V(G)$ such that every component of $G[V_n]$ meets $A \cap V_n$. Minimize the energy of $G[V_n]$ with trace g on $A \cap V_n$ and no other cutoff condition. The minimizers h_n converge pointwise to h. Their gradients, extended by zero on edges absent from $G[V_n]$ solely for comparison in the edge Hilbert space, converge strongly to $\mathrm{d}h$. Since the restrictions are induced and the vertex sets exhaust the graph, every fixed edge is eventually included. All these constructions admit measurable versions.*

Proof. Such an exhaustion can be obtained by enumerating the vertices and including a fixed finite path from each enumerated vertex to a selected boundary anchor, together with all previously included paths. Its induced finite graph is boundary anchored. The finite minimizer's energy is at most $\mathcal{E}_{G[V_n]}(h) \le \mathcal{E}_G(h)$. For each fixed vertex, fix one path to a boundary anchor in its global component. Since the induced sets exhaust all vertices, this entire finite path and its anchor belong

to V_n for all sufficiently large n. Inequality (3.2), applied after subtracting the fixed anchor value, bounds that vertex value with a constant independent of n. Every subsequence therefore has a pointwise convergent further subsequence. Its limit has trace g on all of A and energy at most $\mathcal{E}_G(h)$ by Fatou, so uniqueness identifies it with h. The extended edge fields converge coordinatewise and have bounded Hilbert norms, hence converge weakly; comparison with the restriction of h and lower semicontinuity give convergence of norms and therefore strong convergence.

Measurably enumerate vertices, choose the first available anchor and finite path in the same component, and use the finite path unions just described. Reachability and the selections use countably many finite paths. Each finite Dirichlet system is nonsingular because every finite component has prescribed boundary. Its solution is measurable in its conductances and data. Pointwise limits give the claimed measurable minimizer. No connected exhaustion of a disconnected graph is postulated, and zero extension of an edge field is not a zero boundary condition on vertex values. □

Lemma 3.4 (Every spatial component meets the boundary)**.** *For every bounded closed axis-parallel rectangle Q with nonempty interior, (G_Q, A_Q) is boundary anchored. If $u \in \mathcal{V}_Q$, its extension by zero to $\mathcal{H}$ has exactly the energy $\mathcal{E}_{G_Q}(u)$. Moreover $b|_{G_Q}$ has finite energy. These assertions do not require* (AE-LC).

Proof. A cell meeting Q but not ∂Q is wholly contained in Q°: a connected set meeting the inside and the outside of the closed rectangle must meet its boundary. Any neighbor of this unpinned cell also meets Q, since adjacent cells intersect. Hence every edge from G_Q to its complement has its G_Q endpoint in A_Q.

Suppose a component C of G_Q did not meet A_Q. All its cells would be inside Q°. They have no graph neighbors outside G_Q, and any neighbor inside G_Q belongs to the same component. Thus C has no edge to its complement in the global graph. Global graph connectedness forces $C = \mathcal{H}$, contradicting that the cells cover the unbounded plane. Consequently every component meets A_Q. This argument works for every rectangle, including rectangles whose side offsets are exceptional in (AE-LC).

The crossing-edge observation proves the zero-extension assertion, because the extension vanishes at both endpoints of each crossing edge. Finally, the centroid estimate and Corollary 2.4 imply

$$\mathcal{E}_{G_Q}(b) \le 2 \sum_{H \in \mathcal{H}(Q)} d_H^2 \pi(H) < \infty. \tag{3.3}$$

□

If $Q \subset S^\circ$, a full-energy minimizer on G_S is orthogonal to every $u \in \mathcal{V}_Q$ extended by zero. Its extension vanishes on A_S, and no extra edge energy is introduced by Lemma 3.4. Thus it is an admissible test on G_S. Neither patch needs to be connected, and neither spatial boundary needs to be finite.

Remark 3.5 (Free conditions are not killing conditions)**.** On a ray with vertices $n \ge 0$ and conductances $c(n, n+1) = 2^n$, the full-energy minimizer with value 1 at zero is the constant 1, of energy zero. Prescribing in addition a zero value at the end changes the answer to $f(n) = 2^{-n}$, of energy 1/2: Cauchy–Schwarz gives

$$1 \le \Big(\sum_{n \ge 0} 2^{-n} \Big) \mathcal{E}(f) = 2\mathcal{E}(f).$$

The variations above include all finite-energy zero-trace functions, not just the closure of finitely supported variations. This example illustrates the boundary condition; it is not asserted to satisfy the planar cell hypotheses.

4 Blockwise harmonic coordinates and their specific energy

Fix an independent uniform dyadic system $\mathcal{D}$. For $m \in \mathbb{N}$ form $\mathcal{S}_m$ from (2.5) and let

$$\mathcal{B}_m = \left\{ H : H \cap \bigcup_{S \in \mathcal{S}_m} \partial S \neq \varnothing \right\}. \tag{4.1}$$

Set $\phi_0(H) = b_H$. On each G_S, $S \in \mathcal{S}_m$, let ϕ_m be the full energy minimizer with trace b on A_S. The definitions agree on overlaps: a cell meeting distinct selected blocks meets their boundaries and receives the value b_H. A cell not in $\mathcal{B}_m$ is contained in the union of the selected open interiors, because the closed selected squares cover the plane. Those interiors are disjoint open sets; connectedness of the cell forces it to lie in exactly one. This remains valid when block boundaries accumulate and their union is not closed. Thus this defines a measurable global function, by Lemmas 3.2–3.4.

The partitions coarsen with m, so

$$\mathcal{B}_n \subset \mathcal{B}_m \quad (n \geq m), \qquad \phi_n = \phi_m = \phi_0 \quad \text{on } \mathcal{B}_n. \tag{4.2}$$

Minimizer uniqueness and the behavior of energy under scaling give

$$\phi_m^{C(\mathcal{H}-u),C(\mathcal{D}-u)}(C(H-u)) = C\big(\phi_m^{\mathcal{H},\mathcal{D}}(H) - u\big). \tag{4.3}$$

For a translation- and dilation-covariant edge field θ (its values scale linearly) define

$$\rho_\theta(\omega, z) = \frac{1}{2a_{H_z}} \sum_{H' \sim H_z} c(H_z, H')|\theta(H_z, H')|^2, \qquad \|\theta\|_{\mathrm{sp}}^2 = \mathbf{E}[\rho_\theta(\omega, 0)]. \tag{4.4}$$

The corresponding bilinear form is $\langle \theta, \eta \rangle_{\mathrm{sp}}$. All fields here are gradients, with either choice of orientation; changing orientation changes both signs and not the energy. Put $g_m = \mathrm{d}\phi_m$ and $e_m = \|g_m\|_{\mathrm{sp}}^2$.

Lemma 4.1 (The base embedding has finite specific energy)**.** *With $M_\pi = \mathbf{E}[d_{H_0}^2 \pi(H_0)/a_{H_0}]$, one has*

$$e_0 \leq 2M_\pi < \infty. \tag{4.5}$$

Proof. By (1.2), the rooted density is at most

$$\frac{d_{H_0}^2 \pi(H_0)}{a_{H_0}} + \frac{1}{a_{H_0}} \sum_{H' \sim H_0} c(H_0, H') d_{H'}^2.$$

The two terms have the same expectation by mass transport, using

$$T(\mathcal{H}, w, z) = \frac{c(H_w, H_z) d_{H_z}^2}{a_{H_w} a_{H_z}} \mathbf{1}_{\{H_w \sim H_z\}}.$$

This gives (4.5). □

We use the following energy redistribution identity. For a selected square S, call an edge *active in S* if both endpoints belong to G_S and at least one is not in A_S. The active edge sets for distinct selected squares are disjoint. Indeed an endpoint outside A_S is a cell contained in S°. It cannot meet any other selected square. An adjacent endpoint meets S at a point of that interior cell. If it were an interior cell of a different block, the intersection would have to belong to two disjoint block interiors. Thus another block cannot own the edge. Every edge on which a function vanishing on $\mathcal{B}_m$ has a nonzero gradient has a nonpinned endpoint and is active in exactly one square. An edge meeting several block patches has both endpoints pinned, so its variation gradient is zero; its base energy is never counted with multiplicity in the comparison below.

Lemma 4.2 (Redistribution over blocks)**.** *Suppose a nonnegative covariant coefficient q_e on edges scales quadratically and vanishes unless e is active for the partition $\mathcal{S}_m$. Let S_e be its unique owner block. Then*

$$\mathbf{E}\left[\frac{1}{2a_{H_0}}\sum_{e\ni H_0} q_e\right] = \mathbf{E}\left[\frac{1}{\ell(S_m(0))^2}\sum_{e:S_e=S_m(0)} q_e\right]. \tag{4.6}$$

The same identity holds for signed coefficients when the expected endpoint density of $|q_e|$ is finite.

Proof. For $e=\{H,H'\}$ send its endpoint-spread mass to its block using

$$T(\omega,w,z)=\sum_e \frac{q_e}{2}\left(\frac{\mathbf{1}_{\{w\in H\}}}{a_H}+\frac{\mathbf{1}_{\{w\in H'\}}}{a_{H'}}\right)\frac{\mathbf{1}_{\{z\in S_e\}}}{\ell(S_e)^2}.$$

This has scaling degree -2. Its outgoing and incoming integrals are the two densities in (4.6). Apply marked mass transport and Tonelli. For signed coefficients apply the nonnegative result to the positive and negative parts. □

Proposition 4.3 (Nested energy projections)**.** *For every $m\in\mathbb{N}$, $e_m<\infty$, and*

$$e_m + \|g_m-g_0\|_{\mathrm{sp}}^2 = e_0. \tag{4.7}$$

For $n\ge m$,

$$e_m = e_n + \|g_m-g_n\|_{\mathrm{sp}}^2. \tag{4.8}$$

In particular $e_m\downarrow e_\infty\ge 0$, and the gradients are Cauchy in specific energy.

Proof. On a block $S\in\mathcal{S}_m$, put $v=\phi_m-b$. It vanishes on A_S. Full variational orthogonality gives $\mathcal{E}_{G_S}(\phi_m,v)=0$, and therefore

$$\mathcal{E}_{G_S}(v)=-\mathcal{E}_{G_S}(b,v)\le\left(\sum_{e\text{ active in }S} c(e)|g_0(e)|^2\right)^{1/2}\mathcal{E}_{G_S}(v)^{1/2}.$$

Consequently $\mathcal{E}_{G_S}(v)$ is at most the active-edge base energy. Redistributing both nonnegative energies by Lemma 4.2 shows $\|g_m-g_0\|_{\mathrm{sp}}^2\le e_0$. Thus $e_m<\infty$ by the triangle inequality. Redistribute the signed coefficients $c(e)g_m(e)\cdot(g_m(e)-g_0(e))$. Their absolute endpoint density is integrable by Cauchy–Schwarz. Their sum on every block is zero by orthogonality, so $\langle g_m, g_m-g_0\rangle_{\mathrm{sp}}=0$, proving (4.7).

For later steps note that finite specific energy and Proposition 2.3 imply local spatial energy finiteness of every ϕ_m. Indeed, for each bounded spatial patch all its cells lie in some bounded disk by (2.2); their full endpoint energies are then counted by the integral of ρ_{g_m} over that disk.

If $n\ge m$, $\phi_m-\phi_n$ vanishes on $\mathcal{B}_n$ by (4.2), and has finite energy on each G_S, $S\in\mathcal{S}_n$. The n-block minimizer is orthogonal to this full finite-energy variation. Apply signed redistribution for the n-partition to obtain $\langle g_n, g_m-g_n\rangle_{\mathrm{sp}}=0$. Expanding the squared norm of $g_m=g_n+(g_m-g_n)$ proves (4.8). □

5 The limit: local convergence and free boundary conditions

Lemma 5.1 (Specific-energy convergence controls spatial patches)**.** *For a covariant edge field θ of finite specific energy and every $R<\infty$, almost surely*

$$\sum_{e\in E(\mathcal{H}(\overline{B}_R(0)))} c(e)|\theta(e)|^2\le\int_{\overline{B}_{R+D_R}(0)}\rho_\theta(z)\,dz\le(R+D_R)^2\mathcal{M}(\rho_\theta). \tag{5.1}$$

A zero specific norm implies that θ vanishes on every edge almost surely. In particular, if $\|\theta_j\|_{\mathrm{sp}}^2 \le 2^{-4j}$, then almost surely

$$\mathcal{M}(\rho_{\theta_j}) \le 2^{-2j} \quad \textit{for every sufficiently large } j. \tag{5.2}$$

Proof. Every cell in $\mathcal{H}(\overline{B}_R(0))$ is contained in $\overline{B}_{R+D_R}(0)$. Integrating its endpoint density counts an internal edge twice with the factor $1/2$, so gives its full energy. This proves (5.1). If the specific norm is zero, Proposition 2.3 gives $\mathcal{M}(\rho_\theta) = 0$ almost surely. Equation (5.1), on integer-radius disks, then detects every edge and makes its value zero. Proposition 2.3 bounds the probability of failure of (5.2) by $512\,2^{-2j}$, a summable sequence. Apply Borel–Cantelli. □

Proposition 5.2 (Existence of the limiting potential)**.** *There is a measurable covariant gradient $g = \mathrm{d}\Phi$, with $\Phi(H_0) = 0$, such that*

$$\|g\|_{\mathrm{sp}}^2 = e_\infty, \qquad \|g_m - g\|_{\mathrm{sp}}^2 = e_m - e_\infty \longrightarrow 0. \tag{5.3}$$

The convergence statements in Theorem 1.2(d) hold. Moreover Φ satisfies (1.7) *on every bounded rectangle.*

Proof. Choose a deterministic increasing subsequence m_j with $\|g_{m_{j+1}} - g_{m_j}\|_{\mathrm{sp}}^2 \le 2^{-4j}$, using Proposition 4.3. Lemma 5.1 shows that on each bounded spatial patch the square roots of the energies of the successive differences are summable almost surely. Hence the gradients converge in the edge Hilbert space of every such patch. Intersect these events for integer-radius disks. Every graph edge and every finite graph cycle lies in one such patch. Limits of the finite cycle sums are zero. Thus the edge limit is an exact gradient: define $\Phi(H)$ by integrating it along any finite graph path from H_0 to H. This is independent of the path, and the normalized potentials converge pointwise at all vertices along m_j.

Fatou and (4.8) give

$$\|g_m - g\|_{\mathrm{sp}}^2 \le \liminf_{j\to\infty} \|g_m - g_{m_j}\|_{\mathrm{sp}}^2 = e_m - e_\infty.$$

It follows that $g_m \to g$ in specific energy, so $\|g\|_{\mathrm{sp}}^2 = e_\infty$ and passage to the limit in (4.8) gives equality in (5.3). For the applications of mass transport to g, covariance is needed for an actual measurable version, not just at the original root. Use the set of marked configurations on which the deterministic subsequence converges in energy on every bounded spatial patch. This set is measurable (test integer-radius disks), has probability one, and is invariant under every translation and positive dilation: any transformed bounded patch is contained in an original bounded patch, and gradient norms scale by the corresponding constant. The limiting cycle sums vanish there. Define g by the edge limit on this invariant set and as the zero field off it. This is a measurable, exactly covariant edge field. The normalized potential on the good set is obtained by finite-path integration, so it has the asserted covariance up to the change of root.

For convergence in probability on a finite graph neighborhood, fix its finitely many paths from H_0 and bound potential differences by the finite-path energy inequality. Those paths lie in a random finite-radius disk. By Proposition 2.3 and (5.3), $\mathcal{M}(\rho_{g_m - g}) \to 0$ in probability. Equation (5.1), followed by truncation of the fixed random radius and path-resistance factors, gives the desired convergence.

Fix a bounded rectangle Q. Its cell union is bounded by (2.2), and for every sufficiently large m it is contained in the interior of the selected origin block $S_m(0)$. Each ϕ_m is therefore orthogonal to

every $u \in \mathcal{V}_Q$, extended by zero. Along the almost-sure subsequence the gradients converge strongly on G_Q, so

$$\mathcal{E}_{G_Q}(\Phi, u) = \lim_{j\to\infty} \mathcal{E}_{G_Q}(\phi_{m_j}, u) = 0.$$

Local finite energy of Φ follows from that same convergence. The full-probability local-convergence event is fixed for integer-radius disks; once on it, this argument works for every bounded rectangle and every variation, without an uncountable intersection of exceptional events. Lemma 3.2 identifies the restriction as the unique full-energy minimizer with its own trace. To get discrete harmonicity at a given cell, take a rectangle containing that cell and all its neighbors in its interior and use the single-vertex variation. □

Proposition 5.3 (The auxiliary grid disappears)**.** *The gradient g and the normalized potential Φ of Proposition 5.2 are measurable functions of $\mathcal{H}$ alone.*

Proof. Given $\mathcal{H}$, take independent grids $\mathcal{D}_1, \mathcal{D}_2$ and construct gradients g^1, g^2 and approximants g^1_m, g^2_m. All specific inner products in this proof are on the joint marked law. By Proposition 5.2, Φ^1 satisfies full variational orthogonality on every block of the second grid. The variation $\phi^2_m - b$ vanishes on that partition's boundary cells. Signed redistribution therefore gives

$$\langle g^1, g^2_m - g_0\rangle_{\mathrm{sp}} = 0.$$

The signed coefficients are integrable by Cauchy–Schwarz, and their local sums are defined because both fields have finite energy on every bounded spatial patch. Send $m \to \infty$ to obtain $\langle g^1, g^2 - g_0\rangle_{\mathrm{sp}} = 0$. Also $\langle g^1, g^1 - g_0\rangle_{\mathrm{sp}} = 0$ by the same argument with the first grid. Hence $\langle g^1, g^1 - g^2\rangle_{\mathrm{sp}} = 0$. Interchanging the grids gives $\langle g^2, g^2 - g^1\rangle_{\mathrm{sp}} = 0$, and adding yields

$$\|g^1 - g^2\|^2_{\mathrm{sp}} = 0.$$

A zero specific norm implies equality at every edge almost surely: its nonnegative rooted density has expectation zero; Proposition 2.3 then makes its integral on every disk zero, which detects every edge via a positive-area endpoint cell. Since both potentials are zero at H_0, they agree at every vertex. Conditional on $\mathcal{H}$, two independent copies thus agree almost surely. To make the measurability conclusion without assuming moments of the vertex values, encode the real coordinates by their arctangents, with zero at unused cell labels. For any one coordinate Z, conditional independence gives

$$\mathbf{E}[(Z^1 - Z^2)^2 \mid \mathcal{H}] = 2\,\mathrm{Var}(Z^1 \mid \mathcal{H}) = 0.$$

Thus each bounded coordinate is $\mathcal{H}$-measurable. Inverting arctangent and using the countable cell coding proves that the entire gradient and normalized potential are $\mathcal{H}$-measurable.

There is also an exactly covariant version. Use the similarity-invariant good set of marked configurations in Proposition 5.2. The set of unmarked environments for which almost every grid belongs to that good set and the grid-conditional gradient law is a point mass is itself similarity invariant: a similarity sends a uniform grid to a uniform grid, and the marked gradient transforms covariantly. Take the value of that point mass on this set and the zero edge field elsewhere. The resulting unmarked edge field is measurable and commutes with similarities. Integrating it from the new root gives precisely (1.9). □

6 Uniform sublinearity

6.1 A fixed block approximation is sublinear

Lemma 6.1 (Fixed-parameter blocks are not macroscopic)**.** *For every fixed $m \in \mathbb{N}$,*

$$s_m(R) := \sup\{\ell(S) : S \in \mathcal{S}_m,\ S \cap \overline{B}_R(0) \neq \varnothing\} \quad \textit{satisfies} \quad s_m(R)/R \longrightarrow 0. \tag{6.1}$$

Consequently

$$\frac{1}{R} \sup_{H \in \mathcal{H}(\overline{B}_R(0))} |\phi_m(H) - b_H| \longrightarrow 0 \quad \textit{almost surely.} \tag{6.2}$$

Proof. Fix $\varepsilon > 0$. If S meets $\overline{B}_R(0)$ and $\ell(S) \geq \varepsilon R$, every ancestor T of S lies in a disk of radius

$$R + \sqrt{2}\ell(T) \leq (\varepsilon^{-1} + \sqrt{2})\ell(T)$$

about the origin. Set $K_\varepsilon = \varepsilon^{-1} + \sqrt{2}$ and

$$\epsilon(t) = \sup_{r \geq t} \frac{D_r}{r}.$$

For all sufficiently large t this is finite, and it decreases to zero by (2.2). Since $\ell(T) \geq \varepsilon R$, the preceding containment gives the uniform bound

$$\frac{D(T)}{\ell(T)} \leq K_\varepsilon \epsilon(K_\varepsilon \ell(T)) \leq K_\varepsilon \epsilon(K_\varepsilon \varepsilon R).$$

Taking the supremum over ancestors proves that $a(S) \to 0$ uniformly over the indicated blocks, so $\kappa(S) \geq b(S) \to \infty$. This contradicts $\kappa(S) \leq m$ for a selected block. It proves (6.1).

If H meets $\overline{B}_R(0)$ and is pinned, $\phi_m(H) = b_H$. Otherwise it is contained in the interior of a block S meeting $\overline{B}_R(0)$. For large R, (6.1) places S in $\overline{B}_{2R}(0)$. Every boundary cell of S has diameter at most $D_{2R} = o(R)$, and its centroid is within that distance of S. Apply the coordinatewise maximum principle to the block minimizer. Both b_H and the range of ϕ_m are in the corresponding coordinate enlargement of S, giving a bound by a constant times $s_m(R) + D_{2R} = o(R)$. □

This replaces [1, Lemma 2.18]. Its proof there bounds block size using the number of cells in a block. That argument is not used here.

6.2 The residual has small energy on large spatial patches

Write $r_m = \Phi - \phi_m$; only its gradient matters in energy estimates. By (5.3), $\|\mathrm{d}r_m\|_{\mathrm{sp}}^2 \to 0$. Choose another deterministic subsequence n_j such that

$$\|\mathrm{d}r_{n_j}\|_{\mathrm{sp}}^2 \leq 2^{-4j}. \tag{6.3}$$

Lemma 5.1 and (2.2) give, almost surely,

$$\eta_j := \limsup_{R \to \infty} \frac{1}{R^2} \mathcal{E}_{\mathcal{H}(\overline{B}_R(0))}(r_{n_j}) \leq 4\mathcal{M}(\rho_{\mathrm{d}r_{n_j}}) \longrightarrow 0. \tag{6.4}$$

Every quantity in this display is well-defined even if $\mathcal{H}(\overline{B}_R(0))$ is infinite.

Lemma 6.2 (Energy controls average line variation)**.** *Let $Q_R = [-2R, 2R]^2$ and $L_y = [-2R, 2R] \times \{y\}$, $-2R \le y \le 2R$. For a function f of finite energy on G_{Q_R} define*

$$V_f(y) = \sum_{\{H,H'\} \in E(\mathcal{H}(L_y))} |f(H') - f(H)|.$$

Then

$$\int_{-2R}^{2R} V_f(y)\, dy \le \left(\sum_{H \in \mathcal{H}(Q_R)} d_H^2 \pi^*(H) \right)^{1/2} \mathcal{E}_{G_{Q_R}}(f)^{1/2}. \tag{6.5}$$

The same inequality holds for vertical line segments. For $y \notin N_h$ with $V_f(y) < \infty$, one has $\operatorname{osc}_{\mathcal{H}(L_y)} f \le V_f(y)$. The vertical oscillation assertion requires $x \notin N_v$.

Proof. For an edge $e = \{H, H'\}$ let λ_e be the length of the set of y for which both endpoints meet L_y. This length is at most $\min(d_H, d_{H'})$. Tonelli and Cauchy–Schwarz give

$$\int V_f = \sum_{e \in E(G_{Q_R})} \lambda_e |\mathrm{d}f(e)| \le \left(\sum_e \lambda_e^2 / c(e) \right)^{1/2} \mathcal{E}_{G_{Q_R}}(f)^{1/2}.$$

The first sum is at most $\sum_{H \in \mathcal{H}(Q_R)} d_H^2 \pi^*(H)$. This proves (6.5); the argument allows countably many edges. For $y \notin N_h$, (AE-LC) says that any two cells meeting L_y are joined by a finite path within that subgraph, and the variation along this path is at most the full nonnegative edge sum. This gives the oscillation assertion. The vertical case is identical. □

This is the Cauchy–Schwarz line estimate of [1, Lemma 2.17], proved here for countable edge sets; the oscillation conclusion is used only at nonexceptional offsets. The use of finite *paths*, rather than finiteness of the whole line subgraph, is explicit.

6.3 From good lines to all cells

Proof of (1.10). Work on the common probability-one event where all preceding conclusions hold for the two deterministic subsequences and every positive integer m. From (2.9), choose a finite random K such that for every sufficiently large R,

$$\sum_{H \in \mathcal{H}(Q_R)} d_H^2 \pi^*(H) \le K R^2.$$

Since $Q_R \subset \overline{B}_{3R}(0)$, (6.4) and Lemma 6.2 imply

$$\lim_{j \to \infty} \limsup_{R \to \infty} \frac{1}{R^2} \int_{-2R}^{2R} V_{r_{n_j}}(y)\, dy = 0, \tag{6.6}$$

and the same for vertical lines.

Fix $N \ge 5$ and write $\alpha = 1/N$. After fixing this sample and N, choose one fixed $m = n_j$ so large that, for every sufficiently large R, both integrals in (6.6) are at most $\alpha^2 R^2/4$. This m is not allowed to depend on R. The full-probability event already includes (6.2) for every integer m, so this pathwise choice causes no exceptional-set problem. Markov's inequality shows that the set of bad horizontal coordinates, where $V_{r_m}(y) > \alpha R$, has length at most $\alpha R/4$; the vertical bad set satisfies the same bound.

Divide $[-2R, 2R]$ into $4N$ intervals of length αR. The interior of each interval contains a horizontal coordinate outside both the variation-bad set and N_h, and a vertical coordinate outside both its variation-bad set and N_v. Indeed each interval has length αR, the variation-bad set has length at most $\alpha R/4$, and the additional exceptional set is null. Call these selected coordinates good. Choose one of each in every interval interior. The choices are arbitrary real coordinates made pathwise; they are not required to be rational, since a null exceptional set may contain all rationals. No subsequent expectation or conditional law depends on these choices. The coordinates are distinct, and the resulting horizontal and vertical line segments form a rectangular grid. Consecutive selected coordinates are at distance at most $2\alpha R$, and the extreme lines surround $\overline{B}_R(0)$ with a positive macroscopic margin.

Let $\mathcal{T}_R$ be the union of all cells meeting the selected line segments. On each line r_m has oscillation at most αR. Any horizontal and vertical lines meet at a point of a cell, so their cell subgraphs intersect. Connecting through at most three lines gives

$$\operatorname{osc}_{\mathcal{T}_R} r_m \le 3\alpha R. \tag{6.7}$$

In the rest of the proof, every $o(R)$ term may depend on the fixed sample, N, and this fixed m, but is uniform over all the cells and grid rectangles in question. Put $\chi = \Phi - b$ and $\chi_m = \phi_m - b$. By (6.2), $\sup_{H \in \mathcal{H}(Q_R)} |\chi_m(H)| = o(R)$ for the fixed chosen m. Thus

$$\operatorname{osc}_{\mathcal{T}_R} \chi \le 3\alpha R + o(R). \tag{6.8}$$

Choose one cell $H_* \in \mathcal{T}_R$ and put $c_R = \chi(H_*)$. Every grid cell has $|\chi(H) - c_R| \le 3\alpha R + o(R)$.

Consider a cell H meeting $\overline{B}_R(0)$ which is not in $\mathcal{T}_R$. It is contained in the interior of one grid rectangle Q', because it is connected and cannot leave that rectangle without meeting its boundary. The sides of Q' are at most $2\alpha R$. Every cell H' meeting $\partial Q'$ is in $\mathcal{T}_R$, and

$$|b_H - b_{H'}| \le \operatorname{diam}(Q') + d_H + d_{H'} \le 2\sqrt{2}\alpha R + 2D_{3R}. \tag{6.9}$$

Use the full variational maximum principle for Φ on $G_{Q'}$. Its boundary coordinates, relative to the fixed vector $b_H + c_R$, are bounded by (6.8) and (6.9). Hence, coordinatewise and then in Euclidean norm,

$$|\chi(H) - c_R| \le 12\alpha R + o(R).$$

The same bound holds for grid cells. In particular it holds for H_0, so

$$\sup_{H \in \mathcal{H}(\overline{B}_R(0))} |\chi(H) - \chi(H_0)| \le 24\alpha R + o(R).$$

Since $\chi(H_0) = -b_{H_0}$ is fixed, the limsup in (1.10) is at most $24/N$. Intersect the events for integer N and send $N \to \infty$.

Finally, for any representatives $z_H \in H$, (1.2) gives $\sup_{H \in \mathcal{H}(\overline{B}_R(0))} |z_H - b_H| \le D_R = o(R)$, proving the representative-independent conclusion. □

Theorem 1.2 now follows from Propositions 5.2 and 5.3, (4.5), and the last proof.

7 Avoidance of spatial infinity

In this section $G = (V, E, c)$ is a fixed connected locally finite graph, and X denotes its canonical path-regular reflected walk with an admissible rate w. This X is a fast-clock process; the area-clock notation begins in Section 9. Write $\mathcal{E} = \mathcal{E}_G$ and

$$\sigma_o = \inf\{t > 0 : X_t \ne o\}, \qquad \tau_o^+ = \inf\{t \ge \sigma_o : X_t = o\}.$$

These times are finite and attained at vertices for the regular reflected version. We use the vertex-valued strong Markov property and the finite-target hitting representation from [2], not a strong Markov assertion at an abstract end.

For its spatial application take $V = \mathcal{H}$, representatives $z_H \in H$, and assume the deterministic estimates

$$D(R) := \sup_{H \cap \overline{B}_R(0) \neq \varnothing} d_H = o(R), \qquad W_\pi(R) := \sum_{H \cap \overline{B}_R(0) \neq \varnothing} d_H^2 \pi(H) \le CR^2 \quad (R \ge R_*). \tag{7.1}$$

Here $C, R_* < \infty$ may depend on the fixed environment. Lemma 2.1 and Corollary 2.4 prove these hypotheses under the main assumptions. We abbreviate W_π to W within the annular calculation. Adjacency implies intersection throughout.

$$H \sim H' \implies H \cap H' \neq \varnothing. \tag{7.2}$$

Choose $r_0 > |z_o|$ sufficiently large that $r_0 \ge R_*$ and

$$D(R) \le R/100 \quad (R \ge r_0). \tag{7.3}$$

Theorem 7.1 (Spatial boundedness of the reflected path)**.** *For every starting cell o and every admissible regular reflected clock,*

$$\mathbf{P}_o^{\mathrm{ref}}[\exists t \in [\sigma_o, \tau_o^+) : X_t \in \mathcal{H},\ |z_{X_t}| \ge 2^n r_0] \le \frac{32C}{\pi(o)(\log 2)^2 n} \quad (n \ge 1). \tag{7.4}$$

Almost surely, simultaneously for all finite T,

$$\sup\{|z_{X_t}| : 0 \le t \le T,\ X_t \in \mathcal{H}\} < \infty. \tag{7.5}$$

No end-valued time has a spatial-infinity label in Definition 7.7. This excludes all such labels at once, not just each fixed end separately.

7.1 The finite-target variational identity

We first work on an arbitrary graph G as above, without spatial assumptions. The next lemma records the analytic part in full, so that no infinite-boundary Dirichlet principle is implicit.

Lemma 7.2 (Finite-target minimizer)**.** *Fix $o \in V$ and a nonempty finite set $F \subset V \setminus \{o\}$. There is a unique function $h_F : V \to \mathbb{R}$ minimizing* (1.5) *among all functions satisfying*

$$h_F(o) = 0, \qquad h_F|_F = 1. \tag{7.6}$$

It has finite energy, takes values in $[0,1]$, and satisfies

$$\mathcal{E}(h_F, \eta) = 0 \tag{7.7}$$

for every finite-energy η vanishing on $\{o\} \cup F$.

Proof. The function $u = 1 - \mathbf{1}_{\{o\}}$ satisfies (7.6) and has energy $\pi(o)$. Thus the infimum of the energy is finite. Truncation to $[0,1]$ preserves the boundary data and does not increase energy. Choose a minimizing sequence with all values in $[0,1]$. A diagonal subsequence converges at every vertex to a function h_F with the prescribed data. Fatou's lemma for the nonnegative edge sums shows that h_F attains the infimum.

If h and $\widetilde{h}$ are minimizers, then

$$\mathcal{E}\left(\frac{h+\widetilde{h}}{2}\right) = \frac{\mathcal{E}(h)+\mathcal{E}(\widetilde{h})}{2} - \frac{1}{4}\mathcal{E}(h-\widetilde{h}).$$

The minimizing property implies $\mathcal{E}(h-\widetilde{h})=0$. Since the graph is connected and all conductances are positive, $h-\widetilde{h}$ is constant; its value at o is zero. This proves uniqueness, including among competitors not bounded in $[0,1]$.

For an η as in the statement, $h_F + t\eta$ is an admissible competitor for every $t \in \mathbb{R}$. The identity

$$\mathcal{E}(h_F + t\eta) = \mathcal{E}(h_F) + 2t\mathcal{E}(h_F,\eta) + t^2\mathcal{E}(\eta)$$

is justified by absolute convergence of the mixed sum. Its minimum at $t=0$ gives (7.7). □

This is the special boundary-data case of [2, Proposition 1.3]; the proof above uses diagonal compactness rather than the Hilbert-space construction in that paper. The crucial point is that the minimizing class contains *all* finite-energy competitors with the given finite-set values, including functions that do not vanish at other graph ends.

Lemma 7.3 (Reflected excursion identity)**.** *For F and h_F as in Lemma 7.2, let $\tau_F = \inf\{t \ge 0 : X_t \in F\}$. Then*

$$\mathbf{P}_o^{\mathrm{ref}}[\tau_F < \tau_o^+] = \frac{\mathcal{E}(h_F)}{\pi(o)}. \tag{7.8}$$

Consequently, for any finite-energy f with $f(o)=0$ and $f|_F = 1$,

$$\mathbf{P}_o^{\mathrm{ref}}[\tau_F < \tau_o^+] \le \frac{\mathcal{E}(f)}{\pi(o)}. \tag{7.9}$$

Proof. For $x \in V$, the finite-set hitting identity of [2, Theorem 1.6(vi)], with target $\{o\}\cup F$, identifies $h_F(x)$ with the probability that the reflected walk from x hits F before o (with the prescribed values 0 and 1 when starting on the target set).

At σ_o, the process is at a neighbor of o, and that neighbor has law $c(o,x)/\pi(o)$ by [2, Theorem 1.6(iii)]. Apply the strong Markov property at this vertex-valued stopping time, as justified by [2, Lemma 3.10]. We obtain

$$\mathbf{P}_o^{\mathrm{ref}}[\tau_F < \tau_o^+] = \frac{1}{\pi(o)}\sum_{x:x\sim o} c(o,x)h_F(x). \tag{7.10}$$

For $u = 1 - \mathbf{1}_{\{o\}}$, the difference $u - h_F$ is an admissible test function in (7.7). Hence

$$\begin{aligned}\mathcal{E}(h_F) &= \mathcal{E}(h_F, u)\\ &= \sum_{x:x\sim o} c(o,x)(h_F(o)-h_F(x))(u(o)-u(x))\\ &= \sum_{x:x\sim o} c(o,x)h_F(x).\end{aligned} \tag{7.11}$$

Only edges incident to o contribute to the middle expression. Combining (7.10) and (7.11) proves (7.8). Minimality gives $\mathcal{E}(h_F) \le \mathcal{E}(f)$, proving (7.9). □

Corollary 7.4 (A countable target)**.** *Let $D \subset V\setminus\{o\}$ be any set, finite or infinite. If f has finite energy, $f(o)=0$, and $f|_D = 1$, then*

$$\mathbf{P}_o^{\mathrm{ref}}\Big[\exists t \in [\sigma_o, \tau_o^+) : X_t \in D\Big] \le \frac{\mathcal{E}(f)}{\pi(o)}. \tag{7.12}$$

Proof. The assertion is immediate if D is empty. Otherwise choose nonempty finite sets $F_k \uparrow D$. The events that X visits F_k before τ_o^+ increase to the event in (7.12). Apply (7.9) to each F_k with the same competitor f, then use continuity of probability for increasing events. □

No minimizer with prescribed data on the infinite set D is introduced. In particular, (7.12) does not require an infinite-target hitting law or attainment of the infimum of visiting times of D.

7.2 From vanishing energy to avoidance of ends

Proposition 7.5 (Spatial cutoff criterion)**.** *Let $\rho : V \to [0, \infty)$ be finite at every vertex. Suppose that for a fixed $o \in V$ there are $R_n \uparrow \infty$ and finite-energy functions $f_n : V \to [0, 1]$ such that*

$$f_n(o) = 0, \qquad f_n(x) = 1 \text{ if } \rho(x) \geq R_n, \qquad \mathcal{E}(f_n) \longrightarrow 0. \tag{7.13}$$

Then $\mathbf{P}_o^{\mathrm{ref}}$-almost surely the ρ-values at vertex-valued times are bounded on every bounded time interval.

Proof. Set $D_n = \{x : \rho(x) \geq R_n\}$. This set is disjoint from $\{o\}$ by (7.13). Corollary 7.4 gives

$$\mathbf{P}_o^{\mathrm{ref}}\Big[\exists t \in [\sigma_o, \tau_o^+) : X_t \in D_n\Big] \leq \frac{\mathcal{E}(f_n)}{\pi(o)}. \tag{7.14}$$

If the vertex-valued portion of this excursion has unbounded ρ-values, the event on the left occurs for every n. Its probability is therefore zero. This proves boundedness of each excursion, not a uniform bound over all excursions. The holding interval at o has the constant finite spatial value $\rho(o)$, so the full interval from time 0 to the first return has bounded vertex-valued range in ρ.

For completeness, define successive return and departure times by $a_0 = 0$ and

$$b_k = \inf\{t > a_k : X_t \neq o\}, \qquad a_{k+1} = \inf\{t \geq b_k : X_t = o\} \quad (k \geq 0).$$

Reflected recurrence and the strong Markov property imply that these times are finite. The holding lengths $b_k - a_k$ are independent with common law $\mathrm{Exp}(w(o))$; hence their sum is infinite almost surely. Since

$$a_k \geq \sum_{j=0}^{k-1} (b_j - a_j),$$

we have $a_k \to \infty$. The same argument for the first excursion, applied at the vertex-valued stopping time a_k, shows that the vertex range on every $[b_k, a_{k+1})$ has bounded ρ-values. The countable intersection of these probability-one assertions still has probability one. On that event, every bounded time interval meets only finitely many such excursions and holding intervals, so its vertex-valued range is bounded in ρ. □

Remark 7.6 (Measurability of the visiting events)**.** Whenever X_t is a vertex, it remains at that vertex on a nontrivial interval to the right, by [2, Theorem 1.6(ii)]. A vertex visit strictly before a return time therefore has a rational-time witness before that return. The visiting events above, and unboundedness of their spatial range, can consequently be expressed using countably many times and thresholds. Bounds simultaneously for every finite T follow by first taking positive integer T. This also avoids treating a supremum that need not be attained as an actual vertex visit.

7.2.1 Which ends are excluded?

Fix a finite exhaustion $K_m \uparrow V$. Each end ω determines nested connected components $C_m(\omega)$ of $G \setminus K_m$; this is the description in [2, Lemma 5.2].

Definition 7.7. For a finite-valued spatial function ρ, an end ω is at spatial infinity if

$$\lim_{m\to\infty} \inf_{x\in C_m(\omega)} \rho(x) = \infty. \tag{7.15}$$

Equivalently, for every $R < \infty$ some end neighborhood $C_m(\omega)$ is contained in $\{x : \rho(x) > R\}$. For a cell network, we take $\rho(H) = |z_H|$.

Condition (7.15) is independent of the finite exhaustion, since two finite exhaustions are cofinal. It is equivalent to $\rho(x_j) \to \infty$ for every vertex sequence tending to ω in the end topology. Indeed, failure of (7.15) gives a finite upper bound for those infima; after enlarging that bound by one, there are an $R < \infty$ and choices $x_m \in C_m(\omega)$ with $\rho(x_m) \le R$, whereas the forward implication follows from eventual membership in each fixed end neighborhood. Merely exhibiting one ray of unbounded spatial radius is not enough.

Corollary 7.8 (Simultaneous avoidance of all spatial-infinity ends)**.** *Under the hypotheses of Proposition 7.5, almost surely no time $t \ge 0$ has an end label satisfying* (7.15)*.*

Proof. Work on the probability-one event of Proposition 7.5, the simultaneous end-labeling property of [2, Proposition 5.3], and the event that $X_q \in V$ for every nonnegative rational q. The latter follows by countability from [2, Theorem 1.6(i)].

Suppose a time t has end label ω satisfying (7.15). Choose an integer $T > t$ and a finite bound M for all vertex-valued spatial positions on $[0, T]$. Choose m such that $\rho(x) > M + 1$ for every $x \in C_m(\omega)$. By the end-labeling property, for some $\varepsilon > 0$,

$$X_s \in C_m(\omega) \cup \{\infty\} \quad \text{whenever } |s - t| < \varepsilon,$$

with $s \ge 0$. Choose a rational $q \in (t, \min\{t + \varepsilon, T\})$. Then X_q is a vertex in $C_m(\omega)$, contradicting the bound M.

The contradiction is pathwise for every possible t and ω. There is no union of exceptional events indexed by the possibly uncountable set of ends. □

7.3 Logarithmic cutoffs on a countable cell network

We now prove the estimate needed to apply Proposition 7.5. All sums in this section are nonnegative countable sums. Their bounds do not require spatial local finiteness.

Proposition 7.9 (Annular energy estimate)**.** *In the setting of Theorem 7.1, define*

$$s_H = \max\{r_0, |z_H|\}, \qquad f_n(H) = \min\left\{1, \frac{\log(s_H/r_0)}{n \log 2}\right\}. \tag{7.16}$$

Then $f_n(o) = 0$, $f_n(H) = 1$ if $|z_H| \ge 2^n r_0$, and

$$\mathcal{E}(f_n) \le \frac{32C}{n(\log 2)^2}. \tag{7.17}$$

Proof. The assertions about the values of f_n follow immediately from (7.16). Since $z_H \in H$, the cell H meets $\overline{B}_{s_H}(0)$, so (7.3) gives

$$d_H \le s_H/100 \qquad (H \in \mathcal{H}). \tag{7.18}$$

For adjacent H, H', choose a point of $H \cap H'$. The triangle inequality gives

$$|z_H - z_{H'}| \le d_H + d_{H'}. \tag{7.19}$$

The maps $z \mapsto |z|$ and $r \mapsto \max\{r_0, r\}$ are 1-Lipschitz. Thus

$$|s_H - s_{H'}| \le d_H + d_{H'} \le (s_H + s_{H'})/100. \tag{7.20}$$

In particular,

$$\frac{\max\{s_H, s_{H'}\}}{\min\{s_H, s_{H'}\}} \le \frac{101}{99} < 2. \tag{7.21}$$

For $j = 0, \dots, n-1$, let E_j be the set of unoriented edges on which f_n is nonconstant and for which

$$2^j r_0 \le \min\{s_H, s_{H'}\} < 2^{j+1} r_0.$$

These sets partition all edges on which f_n is nonconstant. Indeed, such an edge has $\min\{s_H, s_{H'}\} < 2^n r_0$, while both quantities are at least r_0. By (7.21), both representatives of an edge in E_j belong to $\overline{B}_{2^{j+2} r_0}(0)$.

For $r, r' \ge r_0$, the definition of the clipped logarithm and the mean value theorem yield

$$\left| \min\left\{1, \frac{\log(r/r_0)}{n \log 2}\right\} - \min\left\{1, \frac{\log(r'/r_0)}{n \log 2}\right\} \right| \le \frac{|r - r'|}{n(\log 2) \min\{r, r'\}}.$$

This bound includes edges crossing the inner or outer cutoff. Consequently, for $\{H, H'\} \in E_j$,

$$|f_n(H) - f_n(H')| \le \frac{d_H + d_{H'}}{n(\log 2) 2^j r_0}. \tag{7.22}$$

Squaring, multiplying by conductance, and using $(a+b)^2 \le 2(a^2 + b^2)$ gives

$$\begin{aligned} &\sum_{\{H,H'\} \in E_j} c(H, H') |f_n(H) - f_n(H')|^2 \\ &\quad \le \frac{2}{n^2 (\log 2)^2 \, 4^j r_0^2} \sum_{\{H,H'\} \in E_j} c(H, H')(d_H^2 + d_{H'}^2) \\ &\quad \le \frac{2W(2^{j+2} r_0)}{n^2 (\log 2)^2 \, 4^j r_0^2} \\ &\quad \le \frac{32C}{n^2 (\log 2)^2}. \end{aligned} \tag{7.23}$$

For the second inequality, regroup by incidences at cells: the total coefficient of d_H^2 is at most $\pi(H)$, and each endpoint cell meets the indicated ball. Tonelli's theorem justifies this regrouping even with infinitely many edges in the annulus. The last inequality uses (7.1). Summing over the n groups proves (7.17). □

Proof of Theorem 7.1. Apply Corollary 7.4 with $D = \{H : |z_H| \ge 2^n r_0\}$ and the function in (7.16). Proposition 7.9 gives (7.4). These functions satisfy the hypotheses of Proposition 7.5, so (7.5) follows. Corollary 7.8 gives the end-avoidance assertion. The starting cell was arbitrary, and the proof used only the properties of the reflected process that hold for every admissible rate function. □

Remark 7.10 (Independence of the choice of representatives)**.** Under (7.3), if $\rho(H) \geq r_0$, then

$$\operatorname{dist}(0, H) \geq |z_H| - d_H \geq \frac{99}{100}\rho(H).$$

Since $\operatorname{dist}(0, H) \leq \rho(H)$, tending to spatial infinity along end neighborhoods is equivalent to

$$\lim_{m\to\infty} \inf_{H\in C_m(\omega)} \operatorname{dist}(0, H) = \infty.$$

Thus the class of ends excluded by the theorem does not depend on the representatives. Moreover, (7.18) and (7.5) imply that the union of all cells visited on a bounded time interval is itself spatially bounded, not merely the set of their representatives.

Corollary 7.11 (Quenched formulation)**.** *Suppose a random spatial cell network satisfies the hypotheses of Theorem 7.1 almost surely, with possibly environment-dependent C and R_*. Assume a measurable labeling of its cells and conditional reflected-walk laws given the environment, with an admissible clock. Conditional on almost every environment, the reflected walk from each starting cell satisfies* (7.5) *and never hits an end at spatial infinity.*

Proof. Apply the deterministic theorem to every environment in the probability-one set where its assumptions hold. There are only countably many possible starting cells in each environment. □

8 Process conventions and spatial inputs

We now fix an environment satisfying all the almost-sure spatial conclusions proved above. Choose $z_H \in H$ as in the statement. For reference, the proved estimates are

$$D_R := \sup_{H\in\mathcal{H}(\overline{B}_R(0))} d_H < \infty, \qquad D_R/R \longrightarrow 0, \tag{8.1}$$

$$W(R) := \sum_{H\in\mathcal{H}(\overline{B}_R(0))} d_H^2(\pi(H) + \pi^*(H)) < \infty, \qquad \limsup_{R\to\infty} R^{-2}W(R) < \infty. \tag{8.2}$$

Theorem 1.2 supplies a measurable, unmarked potential, normalized by $\Phi(H_0) = 0$, with

$$\mathbf{E}\left[\frac{1}{2a_{H_0}} \sum_{H\sim H_0} c(H_0, H)|\Phi(H) - \Phi(H_0)|^2\right] < \infty, \qquad \sup_{H\in\mathcal{H}(\overline{B}_R(0))} |\Phi(H) - z_H| = o(R). \tag{8.3}$$

On every bounded spatial rectangle, including those with disconnected restrictions,

$$\mathcal{E}_{\mathcal{H}(Q)}(\Phi, u) = 0 \quad (u|_{\mathcal{H}(\partial Q)} = 0,\ \mathcal{E}_{\mathcal{H}(Q)}(u) < \infty), \qquad \mathcal{E}_{\mathcal{H}(Q)}(\Phi) < \infty. \tag{8.4}$$

The variations are not required to have finite support. The potential is also bounded on every bounded spatial patch by the uniform sublinearity estimate at a larger radius. Theorem 7.1 supplies for the admissible fast process Y

$$\sup\{|z_{Y_s}| : 0 \leq s \leq T,\ Y_s \in \mathcal{H}\} < \infty \quad \text{almost surely for every } T < \infty. \tag{8.5}$$

All ingredients in these displays have been proved in this manuscript. We continue to use spatial mass transport (1.3); when referenced below as the process environment identity it is the same assumption.

$$\mathbf{E}\int_{\mathbb{C}} T(\mathcal{H}, 0, z)\, dz = \mathbf{E}\int_{\mathbb{C}} T(\mathcal{H}, z, 0)\, dz. \tag{8.6}$$

Vertex-function integrands are always set to zero at nonvertex times. Their denominators are evaluated only at vertices. All stochastic processes use the canonical path-regular reflected version, not arbitrary fixed-time modifications. Filtrations have the usual augmentation.

9 A regenerative occupation identity

In this part the environment is fixed in the probability-one set described in Section 8. Write $V = \mathcal{H}$. Choose an admissible fast rate w_0 so large that

$$m_0(H) := \pi(H)/w_0(H) > 0, \qquad \sum_H m_0(H) < \infty.$$

Such a choice is possible by increasing the rates in Gwynne–Sung Theorem 1.6. Let Y be that reflected walk. It is reversible with speed measure m_0, and its form is

$$\mathcal{E}(f) = \sum_{\{H,H'\}} c(H,H')|f(H') - f(H)|^2, \qquad \mathcal{F}_0 = \{f \in \ell^2(m_0) : \mathcal{E}(f) < \infty\}. \tag{9.1}$$

This identification is Gwynne–Sung Proposition 3.15. The extended Dirichlet space is the full finite-energy space. Indeed, if f has finite energy, its bounded clippings $f_n = (-n) \vee (f \wedge n)$ lie in $\mathcal{F}_0$ because $m_0(V) < \infty$. They converge pointwise, and $\mathcal{E}(f_n - f) \to 0$ by dominated convergence: the difference of the unclipped residual across an edge is bounded by $|f(H') - f(H)|$ and tends to zero. Conversely, pointwise limits of energy-Cauchy form-domain functions have finite energy with the limiting gradient, by Fatou and completeness of the edge space. Constants are allowed.

For a starting vertex o, set

$$\sigma_o = \inf\{s > 0 : Y_s \neq o\}, \qquad T_o^+ = \inf\{s \geq \sigma_o : Y_s = o\}.$$

Both times are finite almost surely. The interval $[0, T_o^+)$ includes the initial holding interval at o.

Lemma 9.1 (Occupation during a complete return cycle)**.** *For every nonnegative function f on V,*

$$\mathbf{E}_o \int_0^{T_o^+} f(Y_s)\, ds = \frac{1}{\pi(o)} \sum_{H \in V} m_0(H) f(H). \tag{9.2}$$

Both sides are allowed to be infinite. The integrand is zero at end-valued times.

Proof. For $H = o$, the left side with $f = \mathbf{1}_{\{o\}}$ is the mean initial holding time $m_0(o)/\pi(o)$. For $H \neq o$, kill the process when it reaches o and put

$$K_o(x,y) = \frac{1}{m_0(y)} \mathbf{E}_x \int_0^{\tau_o} \mathbf{1}_{\{Y_s = y\}}\, ds, \qquad x, y \neq o.$$

The killed semigroup is symmetric without using the regular realization constructed later. Put $D = V \setminus \{o\}$ and let P_t be the fast symmetric semigroup. For $t > 0$ the operators

$$\left(\mathbf{1}_D P_{t/2^n} \mathbf{1}_D\right)^{2^n}$$

are self-adjoint. Their kernels decrease to the killed kernel as the dyadic observation mesh is refined: every visit to o contains a positive holding interval and is eventually detected, including a visit at the terminal observation time. Passage to the limit preserves detailed balance. Integration in time and Tonelli give $K_o(x,y) = K_o(y,x)$, even before finiteness of the entries is known.

A visit to o from elsewhere occurs by a jump from one of its finitely many neighbors. This is the vertex left-limit assertion in Gwynne–Sung Lemma 3.13. Compensating these ordinary jumps before the first hit of o gives

$$\sum_{x \sim o} c(o,x) K_o(y,x) = \mathbf{P}_y(\tau_o < \infty) = 1. \tag{9.3}$$

For clarity, compensation here is only at vertices: conditional on a holding interval at x, departure has rate $\pi(x)/m_0(x)$ and the departing neighbor has probability $c(x,o)/\pi(x)$. Summing these conditional intensities over the countably many holding intervals and stopping at τ_o gives the expected number of arrivals to o, which is one. For a rigorous localization, restrict to a bounded fast-time interval and the finite neighbor set of o. Each of these vertices has only finitely many holding intervals in that time window by Gwynne–Sung Lemma 3.11. Apply the vertex holding-time calculation to these counting processes, then let the time bound increase. This uses per-vertex visits, not a global chronological enumeration of all jumps. Monotone convergence gives the displayed identity. No compensation rule at graph ends is used.

After the first jump from o, the strong Markov property yields

$$\begin{aligned}\mathbf{E}_o \int_0^{T_o^+} \mathbf{1}_{\{Y_s=y\}}\, ds &= \frac{m_0(y)}{\pi(o)} \sum_{x\sim o} c(o,x) K_o(x,y) \\ &= \frac{m_0(y)}{\pi(o)}.\end{aligned}$$

The last equality uses symmetry and (9.3). Sum over y with weights $f(y)$ using Tonelli. □

Corollary 9.2 (Expected number of visits)**.** *Let N_H count the holding intervals at H in $[0, T_o^+)$, including the first interval if $H = o$. Then*

$$\mathbf{E}_o N_H = \pi(H)/\pi(o). \tag{9.4}$$

Proof. At each arrival to H before the return, the following complete holding interval has conditional mean $m_0(H)/\pi(H)$. The event that this arrival occurs before the return is known at the arrival time. The strong Markov property and Tonelli therefore identify its total expected occupation as $(m_0(H)/\pi(H))\mathbf{E}_o N_H$. Apply Lemma 9.1. The fixed-vertex local finiteness of holding intervals is part of Gwynne–Sung Lemma 3.11. □

10 Finiteness, divergence, and the time-change form

Lemma 10.1 (Area is locally summable)**.** *For every $R < \infty$,*

$$\sum_{H\in\mathcal{H}(\overline{B}_R(0))} a_H \le \pi(R + D_R)^2 < \infty. \tag{10.1}$$

The π on the right is the numerical constant.

Proof. Every cell meeting $\overline{B}_R(0)$ lies in $\overline{B}_{R+D_R}(0)$. Their interiors are disjoint and their boundaries have zero area. Countable additivity proves the assertion. □

Define

$$A(s) = \int_0^s \frac{a_{Y_u}}{m_0(Y_u)} \mathbf{1}_{\{Y_u\in V\}}\, du. \tag{10.2}$$

Theorem 10.2 (The area clock is a homeomorphism)**.** *Almost surely, $A(s) < \infty$ for every finite s, A is continuous and strictly increasing, and $A(s) \to \infty$ as $s \to \infty$. Thus*

$$X_t = Y_{A^{-1}(t)}, \qquad t \ge 0, \tag{10.3}$$

is defined for all times. It preserves the ordered reflected path and every end label. At a vertex H, its holding time is exponential with mean $a_H/\pi(H)$ and its departing neighbor has probabilities $c(H,H')/\pi(H)$.

Proof. For a complete return cycle from o, localize (10.2) to cells in $\mathcal{H}(\overline{B}_R(0))$. Its expectation, by Lemma 9.1, is

$$\frac{1}{\pi(o)} \sum_{H \in \mathcal{H}(\overline{B}_R(0))} a_H < \infty.$$

Consequently all these localized clocks, for positive integer R, are finite almost surely. On the same event the cycle has bounded spatial vertex range: its duration T_o^+ is finite, and (8.5) holds on every bounded fast-time interval. For some sample-dependent integer R every vertex visited in the cycle belongs to $\mathcal{H}(\overline{B}_R(0))$. Its entire area-clock duration is therefore finite. This uses a countable family of integrability statements before choosing the random radius.

Apply this argument to successive returns to o. Their times tend to infinity, because the independent positive exponential holding times at o have divergent sum. Every bounded fast-time interval meets only finitely many complete return cycles and one terminal partial cycle. Thus A is finite and absolutely continuous on every compact interval. The fast process is vertex-valued at Lebesgue-almost every time; hence the integrand in (10.2) is strictly positive almost everywhere. This proves strict increase.

Each holding interval at o is multiplied in length by $a_o/m_0(o)$. The resulting lengths are independent exponentials of mean $a_o/\pi(o) > 0$, and have infinite total sum. Thus $A(s) \to \infty$. On any other vertex holding interval the same deterministic multiplication gives the stated law, without changing the next-vertex choice. □

Proposition 10.3 (Canonical area-clock law)**.** *The process X is reversible with speed measure $a = \sum_H a_H \delta_H$ and has form*

$$\mathcal{F}_a = \{f \in \ell^2(a) : \mathcal{E}(f) < \infty\}, \qquad \mathcal{E}_a(f,g) = \mathcal{E}(f,g). \tag{10.4}$$

Its vertex transition semigroup is independent of the admissible fast clock used to construct it. It is conservative, is at a vertex almost surely at each fixed time, and is recurrent at every vertex. Under $H \mapsto C(H-u)$ its law is transformed by spatial dilation C and time dilation C^2.

Proof. The integrand in (10.2) is a/m_0. The positive continuous additive functional A has Revuz measure a: this follows from the defining occupation integral and symmetry, or from the density case of the Revuz correspondence. The preceding theorem proves directly that this functional is finite on all bounded path intervals and has full lifetime. Its inverse is a homeomorphism, not an erasure of a time interval.

There is a reference-measure issue worth making explicit. The measure a can be infinite on the compact regular realization used below, so it must not be called Radon there. Lemma 12.3, proved using only the fast form and (8.5), provides an open locally compact realization E_{sp} on which the fast process stays, the fast part form is still (9.1), and a is Radon, smooth, and has full quasi-support. Thus the hypotheses of the full-quasi-support time-change theorem are satisfied on that realization. Apply [3, Theorem 6.2.1], equivalently the full-support case of [4, (2.31)], to obtain (10.4). The extended space of (9.1) is the full finite-energy space, so no boundary restriction is introduced. This forward reference has no dependence on the area-clock form or on a martingale property for Φ.

The closed form (10.4) determines its symmetric semigroup uniquely. This proves independence from m_0. The homeomorphism gives conservativity and recurrence from those of Y. The measure a is carried by the vertices, and every vertex has positive mass. The indicator of the complement of V is zero as an $\ell^2(a)$ element, so its semigroup image is zero a-almost everywhere and hence at each positive-mass vertex. Thus the probability of being at a nonvertex state at any fixed deterministic

time is zero from every vertex start. This assertion is not simultaneous over all times: end-valued random times remain permitted. The image of the fast end-time set also has zero Lebesgue measure: A is absolutely continuous on compact intervals.

Dilation multiplies a by C^2 and leaves conductances unchanged. In the associated generator this divides time rates by C^2. The domain and energy in (10.4), and hence their semigroup, transform accordingly. Translations only relabel cells. This proves the scaling rule without requiring a covariant choice of the auxiliary fast rates. □

Measurability of the conditional law. All laws are taken on the countable cell coding used in the spatial part of this manuscript. One can verify measurability of the area semigroup without making a measurable choice of its regular realization. For finite-support f, minimize

$$\mathcal{E}_{G[V_n]}(u) + \alpha \sum_{H \in V_n} a_H u(H)^2 - 2 \sum_{H \in V_n} a_H f(H) u(H)$$

on a connected finite induced exhaustion, with free cutoff boundary. Uniform energy and $\ell^2(a)$ bounds, diagonal compactness, and Fatou show that every limit is the unique full-form resolvent minimizer. Comparison with restrictions of an arbitrary global competitor identifies that minimizer. Finite problems are measurable linear systems, so their pointwise limit is measurable. For completeness, the comparison step is as follows. If u_n is the finite minimizer, completing the square in the $\ell^2(a)$ term gives uniform bounds on $\mathcal{E}_{G[V_n]}(u_n)$ and $\|u_n\|_{\ell^2(a|_{V_n})}$. Each fixed vertex value is bounded because $a_H > 0$. Any pointwise subsequential limit u lies in the full space in (10.4) by Fatou. The linear term converges because f has finite support. For every global competitor v in that space, finite minimality gives $J_n(u_n) \le J_n(v|_{V_n})$, whose right side tends to $J(v)$ by monotone convergence of the nonnegative quadratic terms. Thus $J(u) \le J(v)$ for all v. Strict convexity for $\alpha > 0$ identifies u uniquely; every subsequence has that limit. This argument neither imposes zero values at the finite cutoff nor replaces the full domain by the closure of finite-support functions. For example, the Euler resolvent formula recovers $P_t f$ as the strong $\ell^2(a)$ limit of $((n/t)R_{n/t})^n f$. Positive atomic weights turn strong convergence into convergence at each vertex, so this also gives a measurable version of each semigroup entry. The semigroup and the countable-coordinate path construction determine its finite-dimensional and path laws.

For exact covariance in temporal mass transport, take the measurable good spatial domain

$$\mathcal{A} = \{\mathcal{H} : D_n, W(n) < \infty \ (n \in \mathbb{N}),\ D_n/n \to 0,\ \limsup_{n\to\infty} W(n)/n^2 < \infty\}.$$

Monotonicity extends these integer-radius conditions to all radii. Translation is handled by bounded changes of radius, and dilation scales W quadratically, so this domain is similarity invariant and has probability one. On it the avoidance, occupation, and clock construction use only those estimates and the geometric cell conditions. In constructing a regular space solely to identify the area law, omit the Φ-coordinates from the core; the spatial cutoffs already give the open spatial realization and its Radon area measure. Thus this law is defined without circular dependence on a selected corrector. The free-resolvent construction above makes it measurable, and uniqueness of the full form makes its covariance an actual identity. Off $\mathcal{A}$ use the constant trajectory at the distinguished initial vertex as a covariant fallback kernel. Transports for almost-sure quantities are set to zero off their invariant good domains. None of these fallback trajectories is used under the environment law of the theorem.

Corollary 10.4 (Area occupation in a return cycle)**.** *If $\tau_o^+(X)$ is the first return after the initial departure of the area-clock process, then, for every $f : V \to [0, \infty]$,*

$$\mathbf{E}_o \int_0^{\tau_o^+(X)} f(X_t)\, dt = \frac{1}{\pi(o)} \sum_H a_H f(H). \tag{10.5}$$

Proof. The time change maps a complete fast-clock cycle onto the corresponding complete area-clock cycle. Change variables in its occupation integral, inserting the density $a_H/m_0(H)$, and apply Lemma 9.1. All integrands are nonnegative, so Tonelli applies even when the expectation of the total duration is infinite. The factor is $\pi(o)^{-1}$, not a_o^{-1}. □

11 The exact-holding clock

Let $h_H = a_H/\pi(H)$. On each complete vertex holding interval of the fast path, replace its length by h_H, preserving the order of all intervals and mapping linearly within each one. The resulting clock is well-defined even though there can be infinitely many intervals in a bounded period.

Proposition 11.1 (Existence of the exact clock)**.** *This replacement is a continuous increasing homeomorphism of the positive time axis. It produces a path $\overline{X}$ with exactly h_H time units at each visit to H. Both the area-clock and exact-clock paths have bounded spatial vertex range on compact time intervals and avoid all spatial-infinity ends. Neither their complete return-cycle durations nor the total area of the environment are required to have finite expectation.*

Proof. In one return cycle, the expected total exact duration contributed by cells in $\mathcal{H}(\overline{B}_R(0))$ is, by (9.4),

$$\mathbf{E}_o \sum_{H \in \mathcal{H}(\overline{B}_R(0))} h_H N_H = \frac{1}{\pi(o)} \sum_{H \in \mathcal{H}(\overline{B}_R(0))} a_H < \infty.$$

The same random-radius and finite-cycle argument as in Theorem 10.2 proves local finiteness of the clock. Its density on every holding interval is positive. Those intervals cover almost every fast time, so it is strictly increasing; the countable sum of their rescaled lengths defines an absolutely continuous clock on each compact interval. Repeated visits to o contribute the fixed positive amount h_o and make the clock diverge. Spatial boundedness and avoidance are invariant under these homeomorphic time changes. □

The asymptotic equivalence of the exact clock and the exponential area clock is not assumed here. It is proved in Part III after establishing temporal averaging. Also, the exact-clock path is not claimed to be Markov in its vertex state alone.

12 A regular realization and the location of its energy

We continue with a fixed environment. It is important not to put the discrete topology on V and declare the full form regular there: in the transient case finitely supported functions need not be energy dense. We give a regular realization on an enlarged compact metrizable space for the fast form (9.1). It will be used only to prove properties of the original reflected path.

Lemma 12.1 (Bounded spatial cutoffs)**.** *Let $\eta_R : \mathbb{R}^2 \to [0, 1]$ be a Lipschitz function equal to one on $\overline{B}_R(0)$, zero off $B_{2R}(0)$, and chosen increasing with R. Each of*

$$\eta_R(z_H), \qquad \eta_R(z_H) z_H^i, \qquad \eta_R(z_H)\Phi^i(H), \qquad i = 1, 2, \tag{12.1}$$

is bounded and has finite graph energy. These functions belong to $\mathcal{F}_0$ and are square integrable against a. In particular they belong to the space on the right side of (10.4)*, without needing its process identification here.*

Proof. If $|z_H| \le L$, then $H \subset \overline{B}_{L+D_L}(0)$. Every neighbor of such a cell meets $\overline{B}_{L+D_L}(0)$ and is contained in $\overline{B}_{L+D_L+D_{L+D_L}}(0)$. Thus, for any fixed L, the cells with representative in $\overline{B}_L(0)$ and all their neighbors lie in one bounded spatial patch. An edge on which one of (12.1) is nonconstant has an endpoint whose representative is in $\overline{B}_{2R}(0)$, so both endpoints lie in such a patch.

Use $|z_H - z_{H'}| \le d_H + d_{H'}$, the Lipschitz bound for η_R, and (8.2). They bound the energy of the first function. For the other functions use

$$|\eta(H)f(H) - \eta(H')f(H')| \le |\eta(H)|\,|f(H) - f(H')| + |f(H')|\,|\eta(H) - \eta(H')|.$$

The functions z and Φ are bounded on the patch; their patch energies are finite by (8.2) and (8.4). This proves the energy claims. Boundedness gives $\ell^2(m_0)$ because $m_0(V) < \infty$. Their supports have finite a-mass by Lemma 10.1, giving $\ell^2(a)$ as well. □

Lemma 12.2 (Compact regular representation)**.** *There is a compact metrizable space E containing V as a dense open subset of isolated points, on which* (9.1) *is regular and all the functions* (12.1)*, for positive integer R, have continuous representatives. The associated conservative Hunt process has the same vertex laws and ordered reflected path as Y.*

Proof. The graph energy space with norm $(\mathcal{E}(f) + \|f\|^2_{\ell^2(m_0)})^{1/2}$ is separable: its image in the direct sum of the countable vertex and edge ℓ^2 spaces is a closed subspace. Bounded truncations are dense in it, by dominated convergence in the vertex and edge sums. Choose a countable energy-dense family of bounded finite-energy functions. Include the constant function 1, every vertex indicator, and the functions in (12.1). Close this family under rational linear combinations, products, and piecewise linear normal contractions with rational breakpoints and rational values. This is a countable set of operations, so the generating family remains countable. Stability under all normal contractions is a property of the full form and does not require adding uncountably many coordinate generators. All these functions are bounded and have finite energy.

Embed V by their joint evaluations in a countable product of compact intervals and take its closure E. Vertex indicators make the embedding injective and make each vertex isolated. The real algebra generated by the continuous coordinates separates points of E, contains constants, and is uniformly dense in $C(E)$ by Stone–Weierstrass. Its rational subalgebra is uniformly dense in it, while the countable original core already supplies form-norm density. It is also dense in the form domain in the form norm. The measure m_0, supported on V, is finite Radon and has full support because V is dense. Thus (9.1) is a regular Dirichlet form on E. The regular form–Hunt process correspondence gives the associated process. Since 1 belongs to its domain with zero energy, it is conservative and recurrent; graph connectedness makes the form irreducible. These are the standard recurrence/irreducibility criteria in [3, Chapters 1 and 4].

At a vertex, testing against its indicator identifies the ordinary transition rates $c(H, H')/m_0(H)$. The reflected form identification [2, Proposition 3.15] identifies the semigroup with that of Y. Equivalently, finite-set hitting measures are the full-energy minimizers, and the vertex-path properties identify the process by Gwynne–Sung's uniqueness theorem. Refining the state space at nonvertex times does not change those vertex laws. For equality of the entire canonical vertex-path law, use the countably many vertex indicators. Their paths are càdlàg and determined by rational evaluations, whose joint laws agree by semigroup equality. The original constructed reflected process has the same fixed-vertex holding-interval regularity. Thus the ordered vertex paths agree in law; this does

not identify arbitrary nonregular path modifications. The end label of any such time is the label determined by the nested components of the surrounding vertex path. □

Lemma 12.3 (A spatial realization with Radon area measure)**.** *In the compact realization of Lemma 12.2, put*

$$E_{\mathrm{sp}} = \bigcup_{N\geq 1} \{x \in E : \eta_N(x) > 1/2\}.$$

This is an open locally compact metrizable space containing every vertex. From every vertex, the fast process stays in E_{sp} for all time. Its part form on E_{sp} has the same abstract domain and energy as (9.1)*. The area measure a is Radon and smooth on E_{sp}, and has full quasi-support for this form.*

Proof. For an integer T, spatial boundedness (8.5) gives an integer N such that $\eta_N(Y_q) = 1$ at every rational $q \in [0, T+1]$. The continuous representative of η_N composed with the Hunt path is right-continuous. Hence it equals one at every time in $[0, T]$. This proves that the path stays in E_{sp}, simultaneously for all T. Killing upon exit from this open set therefore leaves all vertex semigroup entries unchanged. Since the reference measure is carried by the vertices, the part semigroup, and hence its closed form, is unchanged. The part form is regular on the open subspace.

The sets $\{\eta_N > 1/2\}$ increase and cover E_{sp}. Every compact $K \subset E_{\mathrm{sp}}$ is contained in one of them. Its a-mass is bounded by the total area of cells with $|z_H| \leq 2N$, which is finite by Lemma 10.1. Thus a is a locally finite Borel measure on this locally compact metrizable space, hence Radon. It is mutually absolutely continuous with m_0, both being carried by the same countable set of positive-mass vertices. It charges no set of zero form capacity. A Radon measure with that property is smooth in the regular-form setting. Finally, if a quasi-continuous form-domain function vanishes a-almost everywhere, it vanishes at every vertex, hence m_0-almost everywhere, and therefore quasi-everywhere. The standard characterization of quasi-support gives full quasi-support. Equivalently, the density a/m_0 is strictly positive and finite at every vertex. No finite total area is asserted. □

Here are the particular standard Dirichlet-form facts used below. For a regular symmetric form, its Beurling–Deny decomposition separates a strongly local energy, a jumping measure, and a killing measure. A bounded function in the form domain has Fukushima decomposition into a finite-energy martingale additive functional and a continuous zero-energy additive functional. The continuous martingale part is determined by the strongly local energy, and the predictable bracket of the jump martingale is determined by the jumping measure and the reference speed. Weak energy harmonicity on an open set implies the stopped martingale property there for bounded form-domain functions. These facts are used in their regular realization, not asserted for the discrete topology; see [3, Chapters 3–5], [4, Section 2], and the harmonicity theorem in [5]. The functions used here are global bounded form-domain functions, so the additional nonlocal integrability issues for arbitrary local-domain functions do not arise.

Proposition 12.4 (No additional energy on the boundary)**.** *On the regular space E, the strongly local and killing parts of* (9.1) *vanish. Its jumping measure is*

$$J = \frac{1}{2} \sum_{\{H,H'\}\in E(G)} c(H,H')\big(\delta_{(H,H')} + \delta_{(H',H)}\big). \tag{12.2}$$

Our convention is $\mathcal{E}_{\mathrm{jump}}(u) = \int (u(x)-u(y))^2 J(dx,dy)$, with no additional factor $1/2$. Thus $2J$ is the ordered-edge intensity measure used in conventions with a prefactor $1/2$. In particular jumps

have both endpoints in V and traverse ordinary edges. For a bounded $u \in \mathcal{F}_0$, the martingale in its Fukushima decomposition has no continuous martingale part and has predictable bracket

$$\int_0^s \frac{1}{m_0(Y_r)} \sum_{H' \sim Y_r} c(Y_r, H')(u(H') - u(Y_r))^2 \mathbf{1}_{\{Y_r \in V\}}\, dr. \tag{12.3}$$

These statements hold from every starting vertex.

Proof. The measure in (12.2) is Radon off the diagonal. Indeed, on a compact subset of $E \times E$ away from the diagonal, finitely many differences of continuous finite-energy core functions separate the two coordinates, with a strictly positive lower bound on at least one difference. Its J-mass is therefore bounded by a finite sum of their graph energies divided by squared positive separation constants. For every core function, integration of its squared difference against J is exactly its full energy in (9.1). The Beurling–Deny uniqueness theorem therefore identifies the jump part as (12.2) and leaves zero local and killing parts.

The jump compensator has rate $c(H, H')/m_0(H)$ at the vertex H, giving (12.3). Its measure on any pair with a nonvertex endpoint is zero. Apply the jump compensation identity first to jumps of metric size exceeding $1/n$ and then to all n to exclude every nontrivial boundary jump. This is stronger than checking only the rate at a vertex. The absence of a strongly local energy gives zero continuous martingale part.

All these Dirichlet-form statements initially hold outside a properly exceptional set. Such a set has zero m_0-mass and therefore contains no vertex, since $m_0(H) > 0$ for each H. Countably many cutoff functions and vertices suffice for the simultaneous assertions used here. □

The last proposition does not say that a generic finite-energy function has no boundary drift. Its zero-energy additive functional can be nonzero. Removing that drift for Φ is the purpose of the full variational test below.

13 Spatial paths and local harmonicity

Proposition 13.1 (Extension at reflection times)**.** *The vertex functions z_{Y_s} and $\Phi(Y_s)$ have càdlàg extensions along the path, unique from their values at vertex times. Each is continuous at nonvertex times; all its jumps are its ordinary edge differences. The same is true after the area time change.*

Proof. For an integer T, (8.5) bounds the vertex positions on $[0, T+1]$. Choose an integer R larger than this bound. On vertex times in this interval, the cutoff functions in (12.1) equal the uncropped coordinates and potential. Their continuous representatives on E evaluated on the Hunt path are càdlàg. Two sufficiently large cutoff choices agree at every rational time in $[0, T+1]$, almost surely, because those times are vertex-valued. Right continuity then gives agreement at every time in $[0, T]$. Thus they define unambiguous local càdlàg paths, consistently as T grows.

By Proposition 12.4 there are no jumps involving nonvertex states. Applying this to the countable coordinate cutoffs proves continuity of these paths at every nonvertex time, simultaneously. Their jumps at vertices have the stated differences. A right-continuous extension is unique because vertex-valued rational times are dense. The homeomorphism in Theorem 10.2 transfers these assertions to the area clock. The extensions are adapted to the completed natural vertex-path filtration: at a nonvertex time, continuity makes their values limits of past rational vertex values, and at a vertex they are specified vertex functions. The same reasoning applies to each cutoff used as a localizer. Thus the localizing exit times are available in this natural filtration as well as in the Hunt realization. No measurable choice of compactifications across environments is needed for the resulting path law:

at nonvertex times the values just constructed are limits of the measurable coordinate values at rational vertex times. The almost-sure càdlàg extension is determined by this countable data; its law in the usual Skorokhod space is therefore a measurable conditional law. □

Meaning of the coordinates at an end. At a nonvertex time, the notation z_{Y_s} or $\Phi(Y_s)$ henceforth denotes the extension along that trajectory. The Hunt realization may distinguish boundary states having the same graph-end label. We do not assume that the graph-end label alone is a Markov state, or that a deterministic map from all graph ends into the plane exists. The strong Markov and decomposition arguments below are applied on the constructed regular state space; their resulting statements concern the same ordered vertex path and its uniquely determined coordinate limits.

Lemma 13.2 (Full spatial tests imply local process harmonicity)**.** *The path $M_s^0 = \Phi(Y_s)$ of Proposition 13.1 is a local martingale. It has no continuous martingale part, and*

$$\langle (M^0)^i, (M^0)^j \rangle_s = \int_0^s \frac{1}{m_0(Y_r)} \sum_{H' \sim Y_r} c(Y_r, H') \Delta\Phi^i \Delta\Phi^j \mathbf{1}_{\{Y_r \in V\}}\, dr. \tag{13.1}$$

Proof. Fix a starting vertex, and take N sufficiently large that it belongs to U_N. Let $U_N = \{\eta_N > 1/2\}$ and $V_N = \{\eta_N > 1/4\}$ in E, where η_N is the continuous representative of the first function in (12.1). Both families increase, and $\overline{U}_N \subset V_N$ with compact closure. Their vertex sets have $|z_H| \le 2N$. All those cells and their neighbors lie in a bounded spatial patch, by the proof of Lemma 12.1. Choose a bounded rectangle Q_N whose interior contains all cells with $|z_H| \le 2N$, with positive margin. Enlarge it if necessary. The cells of $\mathcal{H}(Q_N)$ lie in a bounded patch. Choose an integer R_N so large that $u_N := \eta_{R_N}\Phi^i$ equals Φ^i on every cell in that patch and on every neighbor of a vertex in U_N.

Take any form-domain function v vanishing quasi-everywhere outside V_N. At vertices it vanishes outside $V_N \cap V$, so its restriction to $\mathcal{H}(Q_N)$ vanishes on $\mathcal{H}(\partial Q_N)$. Edges on which $\mathrm{d}v$ can be nonzero have a supported endpoint and lie in the bounded patch. Extending this variation by zero introduces no further nonzero edge terms. Thus the full variational identity (8.4) gives

$$\mathcal{E}(u_N, v) = \mathcal{E}_{\mathcal{H}(Q_N)}(\Phi^i, v) = 0. \tag{13.2}$$

This applies to all part-domain variations, not just finite-support ones. In particular it applies to every test in $C_c(V_N) \cap \mathcal{F}_0$. The bounded global form-domain harmonicity theorem [5, Theorem 2.7 and Remark 2.8(i)] applies on V_N. Its local conclusion can be used on U_N because $\overline{U}_N$ is a compact subset of V_N. It makes $u_N(Y_{s\wedge\tau_{U_N}})$ a martingale. This two-open-set localization is what checks the relatively compact stopping-domain hypothesis in that theorem.

Before exit, $u_N(Y_s) = (M_s^0)^i$. Equality also holds at exit. If exit occurs by an ordinary jump, both endpoints are among the cells where the cutoff equals one. If exit occurs at a nonvertex time, continuity and the limiting equality before exit give the same conclusion. Its left limit is treated in the same way. Therefore $(M^0)^i$ stopped at τ_{U_N} is a martingale. On any fixed bounded fast-time interval, a sufficiently large N has $\eta_N(Y_s) = 1$ at all vertex times and hence at all times by Proposition 13.1. Consequently $\tau_{U_N} \uparrow \infty$.

By uniqueness of the localized Fukushima decomposition, the continuous zero-energy term for $u_N(Y)$ vanishes up to τ_{U_N}: the stopped function is already a martingale, and the difference from its Fukushima martingale is a continuous zero-energy local martingale, hence zero. Its value at exit is also zero by continuity. Thus we are identifying the bracket of the stopped harmonic coordinate with the stopped Fukushima martingale, not merely using vertex harmonicity.

On $U_N \cap V$ all possible neighboring values of u_N equal those of Φ^i. The bracket formula of Proposition 12.4, including the jump at exit, is therefore (13.1) up to that exit. Let $N \to \infty$ and polarize to obtain the cross-bracket formula. The localization also proves finiteness of the displayed integrals on compact time intervals. The martingale has no continuous part by the same localized identification. □

Theorem 13.3 (Area-clock reflected martingale)**.** *The process $M_t = \Phi(X_t)$ is a locally square-integrable local martingale with* (1.12)*. Its optional quadratic covariation is the sum over ordinary jumps,*

$$[M^i, M^j]_t = \sum_{0<s\le t} \Delta M_s^i \Delta M_s^j, \tag{13.3}$$

and there is no additional bracket contribution on the set of reflection times.

Proof. Time change a local martingale by the inverses of the increasing adapted clock A. Localization by the bounded stopped martingales in Lemma 13.2 justifies the usual stopping-time time-change rule. On a vertex holding interval, $dt = (a_H/m_0(H))\,ds$. Changing variables in (13.1) gives the integrand in (1.12). Both clocks assign zero Lebesgue measure to end-valued times. That observation alone would not exclude a singular boundary bracket. The exclusion comes from the zero strongly local energy and the localized Fukushima identification already proved. More explicitly, the fast optional bracket is a countable sum of atoms at ordinary jumps; composing with a continuous increasing homeomorphism maps these atoms to the corresponding area times and creates no continuous component. This proves (13.3). All claims hold for every vertex start by the exceptional-set observation in Proposition 12.4. □

14 Representatives and continuous interpolation

There can be no assertion that the raw step path is continuous at its ordinary jumps. We use the càdlàg path for martingale arguments and an interpolated path for a statement in the separable space $C([0,\infty), \mathbb{R}^2)$.

On each holding interval $[s,t)$ at a vertex H, with following vertex H', linearly interpolate from z_H to $z_{H'}$ over $[s,t]$. At time zero use the forward initial holding interval. At nonvertex times use the limit given by Proposition 13.1. This gives a continuous path, as follows. Near an end-valued time the graph leaves every fixed finite vertex set, by Gwynne–Sung Proposition 5.3 and Lemma 3.11. The visited cells stay in a bounded spatial region. By the spatial finiteness above a positive diameter in Lemma 2.1, their diameters tend to zero near that time. For precision, if times approaching an end time belong to holding intervals $[s_n, t_n)$, then both endpoints approach that end time. Otherwise disjointness would force a single holding interval either to contain the end time or to have it as an endpoint. The former is impossible; the latter is impossible because the endpoints of a complete holding interval are vertex-valued. Thus both its current and its following vertex eventually leave each finite vertex set. Neighbor differences are at most $d_H + d_{H'}$, so the interpolation error tends to zero. Thus the interpolation remains continuous through an accumulation of holding intervals.

The same reasoning proves representative independence at nonvertex times: any two selections in H differ by at most d_H. In particular, although a priori representatives need not extend as continuous functions on a common compactification, they all give the same spatial value at each reflection time along this path. No assumption of a one-point spatial image for every abstract graph end was used.

15 A temporal mass-transport identity

Let $\mathbf{P}_H^v$ be the two-sided law of the area-clock process in a fixed environment H, started at vertex v at time zero. Given v, its forward and time-reversed backward halves are independent copies of the reversible process. The value at a jump time in the reversed half is taken right-continuously. None of the integrals below depends on this convention. A common measurable trajectory coding is obtained by labeling cells by positive integers and using the one-point compactification $\mathbb{N} \cup \{\infty\}$, with all nonvertex states collapsed to ∞. The fixed-finite-vertex-set continuity properties in Gwynne–Sung Lemmas 3.11–3.13 and Proposition 5.3 make this collapsed path càdlàg; homeomorphic time change preserves this fact. Its rational-time values determine it, its holding intervals, and its end labels. Reindexing the cells and scaling time are measurable operations on this coding. We do not claim the collapsed process is strong Markov at ∞; only its path law is being encoded. The sigma-finite path measure

$$\mathbb{Q}_H = \sum_{v \in H} a_v \mathbf{P}_H^v \tag{15.1}$$

is invariant under every deterministic time shift. This follows directly from the Markov cylinder distributions and $a_v p_t(v,u) = a_u p_t(u,v)$; no finite total speed mass is required.

The annealed probability law is obtained by first sampling $\mathcal{H}$ from its given spatial law and then sampling the two-sided process from H_0. We denote this rooted trajectory by Ω. When discussing its functionals, we forget the absolute spatial origin and retain the distinguished current cell. Formally a cell-rooted environment is encoded by translating the centroid of that cell to zero, with the cell still distinguished. It is not necessary for that centroid to belong to the cell. This convention avoids introducing an extra random point inside each holding interval.

Write $\theta_t\Omega$ for time shift by t, with current root X_t, whenever X_t is a vertex. A common spatial dilation C acts on the trajectory by

$$S_C\Omega = (C\mathcal{H}, (CX_{t/C^2})_{t\in\mathbb{R}}).$$

Spatial translations are understood through the rooted encoding. A functional without an auxiliary time grid will be called *unmarked*. Such a functional may depend on the entire two-sided trajectory, not just the current vertex.

Lemma 15.1 (Temporal mass transport)**.** *Suppose $V(\Omega, s, t) \geq 0$ is measurable, covariant under time shifts,*

$$V(\theta_r\Omega, s-r, t-r) = V(\Omega, s, t),$$

and has parabolic scaling degree -2:

$$V(S_C\Omega, C^2 s, C^2 t) = C^{-2} V(\Omega, s, t).$$

Then

$$\mathbf{E}\int_{\mathbb{R}} V(\Omega, 0, t)\, dt = \mathbf{E}\int_{\mathbb{R}} V(\Omega, t, 0)\, dt. \tag{15.2}$$

Values at nonvertex source or target times can be set to zero. The identity also holds after adjoining an independent time dyadic grid, for covariantly marked transports.

Proof. Define a spatial transport, for unique source and target cells, by

$$T(H, w, z) = \frac{1}{a_{H_z}} \mathbf{E}_H^{H_w} \int_{\mathbb{R}} \mathbf{1}_{\{X_t = H_z\}} V(\Omega, 0, t)\, dt. \tag{15.3}$$

Set it to zero at ambiguous source or target roots. Under spatial dilation, a_{H_z} and dt each acquire a factor C^2, whereas V acquires a factor C^{-2}. Thus T has exactly degree -2 and satisfies the covariance needed in (8.6).

Integration in z cancels the denominator and sums the indicator over all target cells. This gives the left side of (15.2). For the incoming integral, fix the environment and put $o = H_0$. Integration over source cells gives

$$\sum_v \frac{a_v}{a_o} \mathbf{E}_H^v \int_{\mathbb{R}} \mathbf{1}_{\{X_t = o\}} V(\Omega, 0, t)\, dt.$$

For each deterministic t, shift the sigma-finite measure (15.1) by t. The last expression equals

$$\mathbf{E}_H^o \int_{\mathbb{R}} V(\Omega, -t, 0)\, dt = \mathbf{E}_H^o \int_{\mathbb{R}} V(\Omega, s, 0)\, ds.$$

Tonelli justifies all exchanges. Apply spatial mass transport to finish. An independent dyadic time grid can first be averaged out. Its law is invariant under time translations and every positive time dilation, so this averaging preserves both covariance rules. □

This is the replacement for the finite-spatial-restriction argument used to obtain the walk re-rooting identity in GMS Section 3.1. Spatial restrictions here can contain infinitely many vertices; they are not used to approximate the temporal law.

16 Time blocks and a reverse martingale

The holding intervals $[s, t)$, including their vertex entrance time and excluding their departure time, form a countable disjoint family $\mathcal{I}$ covering Lebesgue-almost every time. All have positive finite length. The central interval has two independent exponential pieces, its age and residual lifetime, conditional on $\mathcal{H}, H_0$. Let I_s be the interval containing s when defined.

Lemma 16.1 (Holding intervals are submacroscopic)**.** *Almost surely, for every $K > 0$,*

$$\#\{I \in \mathcal{I} : \operatorname{dist}(0, I) \le K|I|\} < \infty. \tag{16.1}$$

Consequently

$$\sup\{|I| : I \cap [-T, T] \neq \varnothing\} = o(T) \quad (T \to \infty), \tag{16.2}$$

and this supremum is finite for every finite T.

Proof. Apply (15.2) to

$$V(\Omega, s, t) = |I_s|^{-1} \mathbf{1}_{\{\operatorname{dist}(t, I_s) \le K|I_s|\}}.$$

The outgoing integral is $1 + 2K$, while the incoming integral counts the intervals in (16.1). Its expectation is therefore $1 + 2K$. Intersect over integer K. An interval meeting $[-T, T]$ with length at least εT belongs to the one finite collection with $K \ge 1/\varepsilon$. This proves (16.2). The same counting argument gives finiteness above any fixed positive interval length inside a bounded time window, and hence finiteness of the supremum. □

Let $\mathcal{D}_t$ be an independent uniform one-dimensional dyadic system. Its logarithmic phase is uniform modulo one, its origin position is uniform at every level, and its successive parents are uniform among the two choices. Almost surely the origin ancestor chain exhausts $\mathbb{R}$. For $J \in \mathcal{D}_t$ set

$$D(J) = \sup_{I \cap J \neq \varnothing} |I|, \quad \alpha(J) = \sup_{k \ge 0} \frac{D(J^{(k)})}{|J^{(k)}|}, \quad \beta(J) = \alpha(J)^{-1}, \quad \kappa(J) = \sum_{k \ge 0} 4^{-k} \beta(J^{(k)}). \tag{16.3}$$

This is the one-dimensional version of the diameter index in Section 2, not a count of holding intervals. From Lemma 16.1 the quantities are finite and positive. For the parent $J^{(1)}$,

$$\beta(J) \le \beta(J^{(1)}) \le 2\beta(J), \quad \tfrac{4}{3}\beta(J) < \kappa(J) \le 2\beta(J), \quad \kappa(J^{(1)}) = 4(\kappa(J) - \beta(J)) > \kappa(J).$$

Along ancestors $\kappa \to \infty$; down a chain through a holding interval I_s, $\kappa(J) \le 2|J|/|I_s| \to 0$. Thus

$$J_m(s) = \text{largest dyadic interval through } s \text{ with } \kappa(J) \le m$$

defines nested covariant partitions for almost every s. The blocks have finite length, exhaust the time axis through the root, and every root dyadic interval occurs on a nonempty parameter interval $[\kappa(J), \kappa(J^{(1)}))$.

Lemma 16.2 (Conditional temporal averaging)**.** *For an integrable unmarked functional $F(\Omega)$ invariant under S_C, put $F_t = F(\theta_t\Omega)$. Then*

$$A_m := \frac{1}{|J_m(0)|}\int_{J_m(0)} F_t\, dt = \mathbf{E}[F \mid \mathcal{G}_m]. \tag{16.4}$$

Here $\mathcal{G}_m$ is the completed sigma-field of events invariant under common parabolic scaling and under time re-rooting within the origin block, with the time grid shifted as well. These sigma-fields decrease with m.

Proof. For any nonnegative scale-invariant marked U, use the transport

$$V(\Omega, \mathcal{D}_t, s, t) = |J_m(s)|^{-1}\mathbf{1}_{\{t\in J_m(s)\}}U(\theta_t\Omega, \mathcal{D}_t - t).$$

The block-partition identity makes its incoming integral $U(\Omega, \mathcal{D}_t)$. Temporal mass transport therefore says that averaging U over the origin block preserves its expectation. The average A_m is invariant under re-rooting within that same block and common scaling, so is $\mathcal{G}_m$-measurable. Test the expectation identity with $U = \mathbf{1}_B F$, first for positive and negative parts, for $B \in \mathcal{G}_m$. This proves (16.4). Nesting gives the decreasing sigma-fields. □

Lemma 16.3 (Complete dyadic-chain convergence)**.** *The averages A_m converge almost surely and in L^1 as $m \to \infty$ through the positive rationals to*

$$L = \mathbf{E}[F \mid \bigcap_{n\ge 1} \mathcal{G}_n].$$

The same limit holds along the full root dyadic ancestor chain. Also F_t is locally integrable almost surely. The limit has a version invariant under every time re-rooting at a vertex time and common parabolic scaling of the trajectory and grid.

Proof. The integer-parameter reverse martingale theorem gives $A_n \to L$ in L^1 and almost surely. For rational $m \ge n$, the tower property yields

$$A_m - L = \mathbf{E}[A_n - L \mid \mathcal{G}_m].$$

Doob's weak-L^1 inequality, applied to finite rational lists in reverse order and then to increasing lists exhausting the rationals, gives

$$\mathbf{P}\left[\sup_{m\in\mathbb{Q},\, m\ge n} |A_m - L| > \delta\right] \le \frac{\mathbf{E}|A_n - L|}{\delta} \longrightarrow 0. \tag{16.5}$$

The events on the left decrease with n, which proves almost-sure convergence over all rational parameters. Every root dyadic interval is selected on an interval containing a rational; the selecting parameters tend to infinity along ancestors. This proves the full-chain assertion.

Apply the conditional averaging identity also to $|F|$. Each rational-parameter block integral is finite almost surely. These blocks exhaust $\mathbb{R}$, so local integrability follows. Finally use the actual block-average formula to define the limit, assigning zero on the invariant event where that limit or the block construction fails. Under a fixed time re-rooting s, all sufficiently large origin blocks contain both 0 and s and are therefore the same intervals before translation. Their averages coincide. Scaling leaves the averages unchanged. This proves the claimed invariant version, simultaneously in the allowed re-rootings, without asserting that the tail sigma-field is trivial. A re-rooting may put the new origin on a boundary of some small grid intervals. That does not affect the definition of the limit: for every fixed pair of time points, sufficiently large common ancestors contain both in their interiors. Use these eventual unique blocks, ignoring the finitely many smaller ambiguous levels. In particular the invariant version applies at vertex entrance times of the regenerative cycles below. □

17 Why the unmarked limit is deterministic

The invariant sigma-field of trajectory plus time grid can contain relative scale information. It is unnecessary, and generally incorrect, to declare it trivial. We instead average out the auxiliary grid before invoking regeneration and environment ergodicity.

Lemma 17.1 (Regenerative invariant functionals)**.** *Fix a connected recurrent reversible area-clock process in one environment. A bounded measurable unmarked functional B of its two-sided trajectory, invariant under all vertex-valued time re-rootings, is almost surely constant under $\mathbf{P}_H^v$ for each v. Its constant does not depend on v. If B is also covariant under spatial translations and parabolic scaling, this constant is a similarity-invariant measurable function of the environment.*

Proof. First fix v. Consider the law obtained by concatenating a doubly infinite sequence of independent complete return cycles from v, with time zero at the entrance to the zeroth cycle. Cycles include their initial holding at v and the subsequent excursion back to v. Each has finite duration, by recurrence, and the time axes in both directions have infinite length because the initial holding times have positive mean. The mean *full* cycle duration may be infinite; it is not used.

Re-rooting at the next cycle entrance shifts this independent cycle sequence by one. An invariant bounded functional is constant by ergodicity of the two-sided independent-sequence shift. This is the usual Bernoulli-shift consequence of independence, proved by approximating events with finite-coordinate cylinders and separating their shifts.

Relate this entrance-rooted cycle law to $\mathbf{P}_H^v$. Its initial holding length L_v is exponential with mean $h_v = a_v/\pi(v)$ and independent of the following excursion. Bias the cycle law by L_v/h_v and choose time zero uniformly inside that holding interval. The age and residual lifetime then have joint density

$$h_v^{-2} \exp[-(s+t)/h_v]\, \mathbf{1}_{\{s,t>0\}}\, ds\, dt,$$

so they are independent exponentials of mean h_v. Here is an explicit check of the excursion reversal used at this point. Put $q(x) = c(x,v)/a_x$ for neighbors of v and zero otherwise on $V \setminus \{v\}$, and let P_t^v be the semigroup killed at v. The departure law is $a_x q(x)/\pi(v)$. It is followed by the killed process, whose arrival intensity at v is q. If e is the excursion after departure, ζ_e its duration, $0 < t_1 < \cdots < t_k < \ell$, and M_A denotes multiplication by $\mathbf{1}_A$, its cylinder measure with duration in $d\ell$ is

$$\frac{1}{\pi(v)} \left\langle q,\, P_{t_1}^v M_{A_1} P_{t_2-t_1}^v \cdots M_{A_k} P_{\ell-t_k}^v q \right\rangle_{\ell^2(a)} d\ell.$$

This follows from the Markov property and compensation of the single arrival at v, as in (9.3). The function q has finite support and is in $\ell^2(a)$, so these inner products are well-defined. Self-adjointness of the killed semigroup and of the multiplication operators reverses their order. This is exactly the cylinder measure for the time-reversed excursion, with observation times $\ell - t_k, \dots, \ell - t_1$. The excursion law is therefore reversal invariant; the usual right-continuous convention changes only jump-time values. These cylinder identities determine the stopped path law. Independence of the cycles and the strong Markov property now give exactly the two independent Markov halves in $\mathbf{P}_H^v$. The biased law is absolutely continuous with respect to the cycle law, and moving the root inside the holding interval leaves B unchanged. Thus $B = b(H, v)$ under $\mathbf{P}_H^v$.

To remove dependence on v, use the shift-invariant sigma-finite measure (15.1). The event $\{X_0 = u,\ B \neq b(H, u)\}$ is null for this measure; its shift by a deterministic $t > 0$ is null as well. Since B is invariant, on $\{X_t = u\}$ under $\mathbf{P}_H^v$ one has $b(H, v) = b(H, u)$. This event has positive probability for every u, v: follow a fixed finite graph path, using positive holding durations with total smaller than t, then stay at u until t. Hence all constants agree. Measurability follows by integrating B against the measurable conditional path law. To apply the environment ergodicity assumption one can use an exactly invariant version. The set of environments for which the conditional law of B is a point mass for every vertex and all these point masses agree is measurable: test zero conditional variance and equality of means on the countable vertex coding. It is preserved by similarities because B is covariant and the conditional area-clock law transforms canonically. On this invariant set take the common mean, and set it equal to zero elsewhere. This is a measurable similarity-invariant version of the common constant. No origin-dependent exceptional set is used to invoke ergodicity. □

Proposition 17.2 (Unmarked temporal ergodic theorem)**.** *If $F(\Omega)$ is integrable, unmarked, and invariant under parabolic scaling, then, almost surely,*

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T F(\theta_t \Omega)\, dt = \mathbf{E}[F]. \tag{17.1}$$

The corresponding backward-time average has the same limit.

Proof. Let L be the limit in Lemma 16.3, which still depends on the independent time grid. For a bounded Borel ψ, define

$$B(\Omega) = \mathbf{E}_{\mathcal{D}_t}[\psi(L(\Omega, \mathcal{D}_t))].$$

The invariant version of L and translation/dilation invariance of the grid law show that B is invariant under time re-rooting and parabolic scaling. Lemma 17.1 makes it a similarity-invariant function of $\mathcal{H}$. Environment ergodicity makes it the constant $\mathbf{E}[\psi(L)]$.

Because F is unmarked, conditioning first on Ω gives

$$\mathbf{E}[F\psi(L)] = \mathbf{E}[F]\mathbf{E}[\psi(L)].$$

On the other hand, L is the reverse conditional expectation limit, so

$$\mathbf{E}[F\psi(L)] = \mathbf{E}[L\psi(L)].$$

Therefore $\mathbf{E}[(L - \mathbf{E}[F])\psi(L)] = 0$ for every bounded ψ. Taking a bounded sign test yields $L = \mathbf{E}[F]$ almost surely. This establishes deterministic limits for the unmarked averages along the independent root dyadic chain, not triviality for arbitrary marked functionals.

It remains to transfer to intervals $[0, T]$. We give a direct argument instead of importing a dyadic-descendant transfer lemma. First let $F \geq 0$ and put $\mu = \mathbf{E}[F]$. Suppose $T^{-1} \int_0^T F_t\, dt > \mu + \varepsilon$

for arbitrarily large T. Local integrability permits a measurable rational witness $T_n \geq n$ on the event of a witness beyond n. These choices depend on Ω alone. Choose $0 < \delta < 1$ so that $(\mu + \varepsilon)/(1 + \delta) > \mu + \varepsilon/2$.

For an independent uniform time grid, conditional on Ω, the probability that an origin dyadic interval contains $[0, T_n]$ and has length in $[T_n, (1 + \delta)T_n]$ is the strictly positive constant

$$p_\delta = \frac{1}{\log 2} \int_1^{1+\delta} \frac{r-1}{r^2}\, dr > 0. \tag{17.2}$$

Indeed the first dyadic length at least T_n is logarithmically uniform between T_n and $2T_n$, and the root position at that length is uniform. On this event its nonnegative average exceeds $\mu + \varepsilon/2$. Thus

$$\begin{aligned} &p_\delta \mathbf{P}[\text{a forward witness exists beyond } n] \\ &\quad \leq \mathbf{P}\left[\begin{array}{l} \text{some root dyadic interval of length at least } n \\ \text{has average greater than } \mu + \varepsilon/2 \end{array} \right]. \end{aligned}$$

The right side tends to zero by the complete-chain convergence. Hence the forward limsup is at most μ.

For bounded $F \geq 0$, apply this upper bound to $K - F$ for a bound K to get the lower bound. For general $F \geq 0$, apply the bounded result to $F \wedge K$ and let $K \uparrow \infty$, obtaining a liminf at least μ. Split an integrable real F into positive and negative parts. The backward argument uses the reflected interval $[-T, 0]$ and is identical. □

18 Quadratic variation, localization, and Lindeberg

Put Γ as in (1.12) and $\Sigma = \mathbf{E}[\Gamma(\mathcal{H}, H_0)]$. The entries of Γ are unmarked functions of the current rooted cell. They are invariant under scaling: the area in the denominator and the squared corrector increment both scale by C^2. Their absolute expectations are finite by (8.3) and $|xy| \leq (x^2 + y^2)/2$.

Lemma 18.1 (Deterministic nondegenerate bracket)**.** *Σ is finite, symmetric and positive definite. For the area-clock martingale from H_0,*

$$T^{-1}\langle M^i, M^j \rangle_T \longrightarrow \Sigma_{ij} \quad \textit{almost surely.} \tag{18.1}$$

Consequently the brackets of $M^\varepsilon_t = \varepsilon M_{t/\varepsilon^2}$ converge to $t\Sigma$ uniformly on compact time intervals in probability, conditionally on almost every environment.

Proof. Apply Proposition 17.2 to every entry of Γ and use Theorem 13.3. Fubini gives the quenched almost-sure assertion. Uniformity in t for the diagonal entries follows by monotonicity and a finite mesh in t; polarize to obtain the cross terms.

If a deterministic vector $e \neq 0$ had $e^{\mathsf{T}}\Sigma e = 0$, the nonnegative rooted specific energy of $e \cdot \Phi$ would have expectation zero. The zero-specific-energy detection lemma in Lemma 5.1, obtained there from spatial mass transport and the maximal inequality, would imply that $e \cdot \mathrm{d}\Phi$ is zero on every edge almost surely. Connectedness and the normalization make $e \cdot \Phi$ identically zero. Choose a cell containing $Re/|e|$. Its representative differs from this point by at most $D_R = o(R)$, while (8.3) makes Φ differ from that representative by $o(R)$. Taking the inner product with e is a contradiction. A singular deterministic symmetric nonnegative matrix has such a deterministic null vector. Thus Σ is positive definite. □

Lemma 18.2 (Coercivity and small potential jumps from a bounded region)**.** *There is a finite environment-dependent K_0 with*

$$|z_H| \le 2|\Phi(H)| + K_0 \qquad (H \in \mathcal{H}). \tag{18.2}$$

For each fixed $R < \infty$,

$$\lim_{\varepsilon \downarrow 0} \sup_{|z_H| \le R/\varepsilon} \sup_{H' \sim H} \varepsilon |\Phi(H') - \Phi(H)| = 0. \tag{18.3}$$

Proof. At a vertex with large $r = |z_H|$, the cell belongs to $\mathcal{H}(\overline{B}_r(0))$ and (8.3) gives $|\Phi(H) - z_H| \le r/2$. On a fixed bounded region, $|z_H|$ itself is bounded. This proves (18.2).

If $|z_H| \le L$, the cell H is contained in $\overline{B}_{L+D_L}(0)$ and a neighbor H' meets that disk. For $L \to \infty$, (8.1) therefore puts both representatives in $\overline{B}_{2L}(0)$, for all sufficiently large L. Hence the supremum of their corrector errors is $o(L)$ and their representative difference is at most $D_L + D_{L+D_L} = o(L)$. Take $L = R/\varepsilon$. The case $R = 0$ follows by containment in a positive fixed R. □

Lemma 18.3 (Quenched compact containment and conditional Lindeberg)**.** *For each $T < \infty$, the random variables*

$$\sup_{0 \le t \le T} |M_t^\varepsilon|, \qquad \sup_{0 \le t \le T} \varepsilon |Z_{t/\varepsilon^2}|$$

are tight under the conditional law given almost every environment. For each $\delta > 0$,

$$\varepsilon^2 \int_0^{T/\varepsilon^2} \frac{1}{a_{X_s}} \sum_{H' \sim X_s} c(X_s, H') |\Phi(H') - \Phi(X_s)|^2 \mathbf{1}_{\{\varepsilon |\Phi(H') - \Phi(X_s)| > \delta\}} \, ds \longrightarrow 0 \tag{18.4}$$

in quenched probability.

Proof. For a vector locally square-integrable martingale, $|M|^2 - \mathrm{tr}\langle M \rangle$ is a local martingale. Its increasing predictable compensator here is continuous. Stopping when that compensator first reaches L, and applying the submartingale maximal inequality after the usual square-integrable localization, gives

$$\mathbf{P}\left[\sup_{t \le T} |M_t^\varepsilon| > K \mid \mathcal{H} \right] \le \frac{L}{K^2} + \mathbf{P}[\mathrm{tr}\langle M^\varepsilon \rangle_T \ge L \mid \mathcal{H}]. \tag{18.5}$$

The initial martingale value is zero. Lemma 18.1 makes the last probabilities small for L sufficiently large; then let K grow. Coercivity (18.2), first at vertex times and then by the spatial extension, gives compact containment for Z.

On the event that $\varepsilon |Z_s| \le R$ for every $s \le T/\varepsilon^2$, the integrand in (18.4) is identically zero for all sufficiently small ε, by (18.3). This controls *every possible neighboring jump* from the states visited in the region, not only the jumps actually realized. Compact containment lets $R \to \infty$ after $\varepsilon \to 0$, proving (18.4). This distinction is essential: a vanishing realized maximum jump alone would not exclude rare large jumps with a nonvanishing compensator. □

Remark 18.4 (A countercheck on the large-jump condition)**.** Convergence of brackets and a vanishing realized maximum jump in probability are not sufficient by themselves. For example, a symmetric compound Poisson martingale with total jump rate $1/n$ and jump sizes $\pm\sqrt{n}$ has predictable bracket t. On every fixed time interval it has no jump with probability tending to one, and converges to the zero path rather than Brownian motion. Its predictable large-jump second-moment integral does not vanish. Equation (18.4) rules out precisely this defect.

19 The quenched invariance principle

Theorem 19.1 (Area-clock Brownian limit)**.** *From the origin-containing cell, conditionally on almost every environment, the laws of M^ε converge to Brownian motion with covariance Σ in the Skorokhod J_1 space. The interpolated spatial paths satisfy* (1.13)*.*

Proof. Lemmas 18.1 and 18.3 verify the predictable-characteristics form of the martingale functional central limit theorem [6, Theorem VIII.3.22]. To check directly an elementary published multidimensional version, we also give a bounded-jump localization reducing to [7, Theorem 2.1(ii)]. That version asks for negligible expected squared maximum jumps, negligible jumps of the predictable brackets, and bracket convergence.

Fix a good environment. By (18.3), choose deterministic numbers $\varepsilon_n \downarrow 0$ such that for $0 < \varepsilon \le \varepsilon_n$ every possible scaled potential jump from $\{|z_H| \le n/\varepsilon\}$ has size at most $1/n$. Choose them strictly decreasing, and let $r_\varepsilon = \max\{n : \varepsilon \le \varepsilon_n\}$ for sufficiently small ε. Thus $r_\varepsilon \to \infty$ and the possible jump bound for that radius tends to zero. Stop M^ε at

$$\tau_\varepsilon = \inf\{t \ge 0 : \varepsilon |Z_{t/\varepsilon^2}| \ge r_\varepsilon\}.$$

Compact containment implies $\mathbf{P}(\tau_\varepsilon \le T \mid \mathcal{H}) \to 0$ for every fixed T. Every jump of the stopped martingale, including its exit jump, has size at most $1/r_\varepsilon$: before exit its pre-jump vertex is in the indicated spatial region, and jumps at nonvertex states have already been excluded. A continuous exit adds no jump. Its predictable bracket is continuous and agrees with the original bracket on $[0, T]$ whenever $\tau_\varepsilon > T$. It therefore converges to $t\Sigma$, while the expected squared maximum jump is at most r_ε^{-2} and the bracket's maximum jump is zero. These are exactly the hypotheses of Whitt's Theorem 2.1(ii). The stopped martingales converge to the required Brownian motion, and the probability of stopping on a fixed compact interval tends to zero, transferring convergence to M^ε itself.

All assertions are quenched. The environment set is fixed using the spatial conclusions, the finite family of bracket averages, and integer time horizons with rational positive jump thresholds; monotonicity extends the latter to all horizons and thresholds. Applying the sequential martingale theorem to an arbitrary sequence $\varepsilon \downarrow 0$ establishes the stated full limit, not merely a selected subsequence.

By compact containment, with probability arbitrarily close to one the spatial path through time T/ε^2 lies in $\overline{B}_{R/\varepsilon}(0)$. On that event (8.3) gives

$$\sup_{t \le T} \varepsilon |Z_{t/\varepsilon^2} - M_{t/\varepsilon^2}| \longrightarrow 0.$$

The assertion extends from vertex times to all times by their right-continuous extensions. Thus the càdlàg spatial process has the same J_1 limit. Its interpolation error is bounded by the largest ordinary spatial jump whose preceding holding interval meets the time interval. Possible following endpoints lie in a bounded enlargement of the same spatial region. The estimate $|z_H - z_{H'}| \le d_H + d_{H'}$, together with (8.1), makes that error $o(1)$ after scaling. Hence the continuous interpolations converge to the same limit. On continuous paths, J_1 convergence to a continuous limit is uniform on compact intervals. This proves (1.13) in C. □

Theorem 19.2 (Exact holding times have the same limit)**.** *The exact-clock process of Proposition 11.1, with the same spatial interpolation, has the limit* (1.13) *with the same matrix Σ.*

Proof. Use the two-sided exponential area process. For each holding interval I, let $H(I)$ be its vertex and let $L(I)$ be its total duration, including both the age and residual pieces for the central

interval. Define

$$V(\Omega) = \frac{h_{X_0}}{L(I_0)}, \qquad C(t) = \int_0^t \frac{h_{X_s}}{L(I_s)}\, ds. \tag{19.1}$$

This functional is unmarked and parabolically scale invariant. Conditional on $\mathcal{H}, H_0$, $L(I_0)$ is a sum of two independent exponentials of mean h_{H_0}, so

$$\mathbf{E}\left[\frac{h_{H_0}}{L(I_0)} \,\middle|\, \mathcal{H}\right] = 1.$$

For example, integrating against the density $\ell h^{-2} e^{-\ell/h}$ gives this equality directly. It follows from Proposition 17.2 that

$$C(t)/t \longrightarrow 1 \quad \text{almost surely.} \tag{19.2}$$

Local finiteness of C follows also from Proposition 11.1; it rescales each complete holding interval to h_H. Its density is positive almost everywhere, and repeated visits give divergence. It is therefore a homeomorphism. From monotonicity and (19.2),

$$\sup_{0\le t\le T} |\varepsilon^2 C^{-1}(t/\varepsilon^2) - t| \longrightarrow 0 \quad \text{almost surely for each } T < \infty.$$

Composition with this asymptotically identity time change preserves the continuous Brownian limit of Theorem 19.1; joint convergence follows since the second coordinate converges in probability to a deterministic function. No adaptedness of C is used in this composition argument. Indeed $L(I_s)$ includes the future remainder of its holding interval, so the exact-clock construction is intentionally not asserted to be an adapted Markov time change.

At the initial time, (19.1) assigns only the residual fraction of the central exact holding interval. Giving the initial vertex a full holding time changes the clock by an amount between zero and h_{H_0}, thereafter constant. This finite initial change vanishes under diffusive time scaling. Both interpolations move linearly over each holding interval, so time change respects them after this initial adjustment. □

Every fixed starting vertex. Fix a good environment and a vertex v. The area or exact reflected walk from v hits H_0 after a finite time. Its initial spatial range up to that time is bounded. The post-hit ordered path has the law from H_0; for the area clock this is the strong Markov property, and for the exact clock it follows by rescaling its underlying vertex-arrival construction. The rescaled initial duration and range tend to zero. This transfers the limit to v. There are countably many vertices, so the conclusion holds for each of them under almost every environment. It does not furnish uniformity when the starting vertex changes with ε.

Other representatives. On a spatially contained path, two choices in each cell differ by at most $D_{R/\varepsilon}$, which is $o(1/\varepsilon)$. Their scaled interpolations differ by the same bound, including at reflection times as proved in Section 14. This proves the representative assertion of Theorem 1.3.

20 Scope, dependence on GMS, and the geometric revision

Theorem 1.2, Theorem 7.1, and the process arguments complete the proof of Theorem 1.3. The result assumes a cell decomposition satisfying Definition 1.1, mass transport, the diameter–conductance moment, and ergodicity modulo translations and positive dilations. It does not construct such cells for a specific supercritical planar map, prove an SLE limit, or prove uniformity for starting cells varying with the scaling parameter.

The exact change from every-line connectivity is localized as follows. Lemma 3.4 supplies boundary anchoring for *every* rectangle using only global graph connectedness, adjacency intersection, and plane coverage. Lemmas 3.2–3.3 solve the full Dirichlet problem on these potentially disconnected restrictions. The projection identities and every subsequent spatial free-variation test therefore remain valid. In Lemma 6.2 the integral estimate does not require line connectivity; only its oscillation conclusion is restricted to nonexceptional offsets. The good-line selection in Section 6 excludes the null sets as well as the small-measure variation-bad sets. No random dyadic boundary is required to avoid the exceptional offsets.

The GMS architecture is retained: energy-minimizing interpolations, a sublinear corrector, and a martingale functional limit theorem. Its cell-counting blocks are replaced by the diameter-based index; finite-dimensional minimization is replaced by the boundary-anchored full-energy projection; deterministic marked-grid averaging is replaced by a maximal inequality and the unmarked temporal argument. The annular recurrence estimate is used with reflected finite-target harmonic measures, not with a finite spatial exhaustion. The only GMS construction imported directly is its similarity-invariant dyadic grid. The reflected process and standard Dirichlet-form and martingale-limit inputs are the external results listed in the statement section, with their application conditions checked in the process sections.

A A correction to the earlier auxiliary-grid averaging claim

An auxiliary-grid averaging assertion sometimes used in this program would state that for every integrable scale-invariant functional $F(\mathcal{H}, \mathcal{D})$,

$$\frac{1}{\ell(S_k)^2} \int_{S_k} F(\mathcal{H} - z, \mathcal{D} - z)\, dz \longrightarrow \mathbf{E}[F]. \tag{A.1}$$

That formulation is too strong when F can depend on the independent grid. This issue is distinct from the line-connectivity weakening.

For a direct example let $\mathcal{H}$ be the unit-square tiling shifted by a uniform point of $[0,1)^2$, with nearest-neighbor adjacency and unit conductances. It satisfies all environment assumptions, mass transport, and ergodicity modulo translations and scalings. Its moment in (FE) is 16. Let $s(\mathcal{D}) \in \mathbb{R}/\mathbb{Z}$ be the logarithmic phase of the side lengths of an independent uniform dyadic system. Define the bounded scale-invariant functional

$$F(\mathcal{H}, \mathcal{D}) = \cos\left(2\pi \left[s(\mathcal{D}) - \tfrac{1}{2}\log_2 a_{H_0}\right]\right).$$

Under common dilation, both terms inside the brackets increase by the same $\log_2 C$ modulo integers. On the randomly shifted unit-square tiling, $a_{H_z} = 1$ at almost every z. Thus every spatial average in (A.1) equals $\cos(2\pi s(\mathcal{D}))$, whereas $\mathbf{E}[F] = 0$. The grid has retained a relative logarithmic phase.

In particular, averaging an invariant function over the independent grid and obtaining a deterministic expectation does not establish joint tail triviality. The present proof uses only the valid conditional-expectation identity (2.6) and finite-list martingale maximal inequalities. It does not replace a conditional invariant average by an unconditional expectation. Corollary 2.4 supplies the energy conclusion required for the recurrence note even without (A.1); with ergodicity its asymptotic bound is again deterministic.

B Measurability conventions

One can encode the cell collection by labeling a cell with the first rational point in a fixed enumeration of $\mathbb{Q}^2$ lying in its interior. Interior disjointness makes the labels distinct. The ambient space is a

countable product of the Hausdorff hyperspace of compact sets (with an empty symbol), adjacency indicators, and conductance coordinates. Similarities relabel the cells after moving the compact sets. We use its standard Borel coding, completed under the relevant probability laws. All geometric hypotheses are assumed on a measurable probability-one event; no assertion about the Borel complexity of the universal plane-coverage condition is needed.

The selections of squares, the countable edge sums, and the maximal function over rational radii are measurable. For a specified rectangle, a measurable boundary-anchored finite exhaustion of G_Q is obtained by joining each of the first finitely many labeled vertices to the first available vertex of A_Q in its own component, using the first available finite graph path. The finite sets are enlarged cumulatively. Lemma 3.3 then gives measurable minimizers. The limits use deterministic subsequences. To interpret the almost-sure statements on every finite graph neighborhood, use graph-distance balls about H_0; they are finite by graph local finiteness. Arbitrary finite vertex sets follow by connectedness. All covariance assertions for potentials are understood modulo an additive constant; the edge-field assertions have no such ambiguity.

C Normalization and scope of verification

Every energy counts unoriented edges once. For the unit-square tiling with nearest-neighbor unit conductances, the area clock has total jump rate 4, coordinate bracket rates 2, and covariance $\Sigma = 2I$. The exact holding time is $1/4$ and gives the same limit. The occupation coefficient of a return cycle is $\pi(o)^{-1}$, not a_o^{-1}.

The complete manuscript is an unrefereed proof, with three documented self-audit-and-repair rounds supplied separately. It is not a Lean-certified proof. In particular the external reflected-form, harmonicity, and martingale-limit theorems are mathematical citations rather than formalized assumptions discharged by a kernel in this artifact. The accompanying formalization plan distinguishes existing source declarations, unverified build claims, and mathematical infrastructure still to be formalized.

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